

Hypergrid

N-Dimensional Distributed Spreadsheet System

API Specification — Version 1.0

Attribute	Value
Document	Hypergrid API Specification v1.0
System	Hypergrid N-Dimensional Distributed Spreadsheet System (NDSS)
Platforms	Apapo · Kogi IW-OS · Ume B-OS · Qala SF-OS · Any platform on Hypergrid substrate
Version	1.0 — March 2026
Status	Authoritative Reference — Active Design
Audience	API consumers · Integration engineers · SDK authors · Platform developers · Domain system builders
Base URL	https://{grid_host}/api/v1
Protocols	REST (JSON over HTTPS/2) · gRPC (internal) · WebSocket (WSS) · Server-Sent Events (SSE)
Auth	Bearer JWT (RS256) · API Key (X-API-Key header) · mTLS (internal gRPC only)
Rate Limiting	Per-identity: 1,000 req/min default. Per-endpoint overrides documented per section.
Idempotency	POST/PUT/PATCH: support X-Idempotency-Key header (stored 24h in Redis).
Versioning	URL path versioning: /api/v1, /api/v2. Breaking changes always bump the major version.
Content-Type	application/json for REST. application/x-ndjson for streaming responses.
Classification	Confidential — Internal Use Only

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Part I — Foundations

1. Overview

The Hypergrid API exposes the complete functionality of the Hypergrid N-Dimensional Distributed Spreadsheet System over HTTPS/2 REST, WebSocket, and gRPC interfaces. The API is the canonical interface for all platform interactions: creating and querying N-dimensional Hypercubes, executing HyperQL queries, managing the Hypergraph, provisioning Spaces and Workspaces, managing identities and namespaces, submitting AI computation requests, and operating federation between Grid deployments.

The API is organized into eleven resource groups: Grids, Hypercubes, HyperCells, HyperQL, Hypergraph, Spaces, Workspaces, Namespaces, Identity, AI, and Federation. Each group has a consistent sub-resource structure and follows the same authentication, error, and pagination conventions.

2. Authentication

2.1 Authentication Methods

Method	Header / Mechanism	Scope	Notes
Bearer JWT	Authorization: Bearer {jwt}	All REST and WebSocket endpoints	RS256 signed JWT. Issued by Hypergrid identity service. Contains: sovereign_entity_id, identity_tag, space_id, scopes, exp. Max lifetime: 24h. Refresh via POST /auth/token/refresh.
API Key	X-API-Key: {key}	All REST endpoints (not WebSocket)	Long-lived. Created via POST /identity/api-keys. Scoped to specific cubes, spaces, or operations. For server-to-server automation.
mTLS	Client certificate in TLS handshake	Internal gRPC only (service mesh)	For inter-service communication only. Not exposed externally. Certificate issued by platform PKI.
Session Token	Cookie: hg_session={token}	Browser clients (claude.ai-style apps)	Short-lived (4h). Automatically refreshed by client SDK. CSRF-protected via SameSite=Strict + CSRF header.

2.2 JWT Token Structure

```
// JWT payload structure (RS256)
{
```

```

"sub": "sovereign_entity_id:uuid",
"iss": "https://grid.hypergrid.io/auth",
"aud": "hypergrid-api",
"exp": 1780000000, // Unix timestamp
"iat": 1779913600,
"jti": "unique-token-id", // for revocation
"hg": {
  "sovereign_entity_id": "uuid",
  "identity_tag": "@alice-dev", // active partition
  "space_id": "uuid", // active Space (optional)
  "workspace_id": "uuid", // active Workspace (optional)
  "permission_tier": 4, // Editor
  "scopes": [
    "grid:read",
    "cubes:read",
    "cubes:write",
    "graph:read",
    "spaces:read",
    "namespaces:read"
  ],
  "cube_restrictions": [], // empty = all cubes
  "grid_id": "uuid" // which grid this token is for
}

```

2.3 Auth Error Codes

HTTP Status	Code	Meaning
401	unauthorized	Missing or malformed token. No Authorization header present or token cannot be parsed.
401	token_expired	JWT exp has passed. Refresh using POST /auth/token/refresh.
401	token_revoked	Token jti has been revoked (logout or key rotation).
403	insufficient_tier	Authenticated but PermissionTier is below the minimum required for this operation.
403	scope_not_granted	The required scope (e.g. cubes:write) is not present in the token's scopes array.
403	cube_restricted	Token has cube_restrictions set and the requested cube is not in the allowed list.
403	isolation_violation	Cross-partition access attempted on a HardIsolated partition without cross-partition scope.
403	policy_denied	A PolicyEngine evaluation blocked this operation. Error body contains policy_id and reason.

3. Base URL and Versioning

```

// Production base URLs
https://{grid_host}/api/v1           // REST API v1
wss://{grid_host}/ws/v1             // WebSocket v1
grpc://{grid_host}:9000             // gRPC (internal only)
https://{grid_host}/api/v1/events   // Server-Sent Events stream

// Grid host examples
kogi.hypergrid.io                   // Kogi production Grid
ume.hypergrid.io                    // Ume production Grid
qala.hypergrid.io                   // Qala production Grid
localhost:8080                      // Local development
grid.company.internal               // Self-hosted deployment

// API version path
/api/v1/{resource}                  // Current stable version
/api/v2/{resource}                  // Next major version (when released)

// Per-grid scoping (when one host serves multiple grids)
/api/v1/grids/{grid_id}/cubes/{id} // Fully qualified
/api/v1/cubes/{id}                  // Implicit: uses grid from JWT

```

4. Common Request / Response Conventions

4.1 Request Headers

Header	Required	Description
Authorization	Yes (JWT)	Bearer {jwt}. Required on all authenticated endpoints.
X-API-Key	Yes (API key)	Mutually exclusive with Authorization. For server-to-server.
Content-Type	Yes (write ops)	application/json for all request bodies.
Accept	No	application/json (default). application/x-ndjson for streaming.
X-Idempotency-Key	No	UUID v4. For POST/PUT/PATCH operations. Prevents duplicate processing. Stored 24h.
X-Request-ID	No	Client-provided request identifier for distributed tracing. Echoed back in response.
X-Grid-ID	No	Override the grid targeted by this request (must match JWT grid_id or federation peer).
X-Identity-Tag	No	Override the active identity partition for this request (must be owned by caller).
X-Space-ID	No	Set the active Space context for this request.
X-Workspace-ID	No	Set the active Workspace context for this request.
If-Match	No	Optimistic concurrency: provide current VectorClock version. Returns 409 if stale.

Header	Required	Description
Prefer	No	return=minimal (no body on success) return=representation (full entity). Default: representation.

4.2 Response Headers

Header	Description
X-Request-ID	Echo of client X-Request-ID or server-generated UUID. Use for support tickets.
X-RateLimit-Limit	Maximum requests per minute for this endpoint.
X-RateLimit-Remaining	Remaining requests in the current rate limit window.
X-RateLimit-Reset	Unix timestamp when the rate limit window resets.
X-Vector-Clock	VectorClock state of the mutated entity after this write operation (JSON-serialized).
X-Entity-Version	Monotonic version counter of the entity after this write.
X-Conflict-ID	Present when a CRDT conflict was recorded during this write (not an error — write still succeeded).
X-Deprecation-Notice	Present when this endpoint or parameter is deprecated. Value = replacement endpoint path.
X-Grid-ID	The Grid that handled this request.
Link	Pagination links: rel=next, rel=prev, rel=first, rel=last with full URL.
ETag	Entity tag for the current representation. Use with If-Match for optimistic concurrency.

4.3 Pagination

All list endpoints use cursor-based pagination for consistency across large datasets. Offset-based pagination is also supported for smaller datasets but is deprecated for large cubes.

```
// Query parameters for all list endpoints
?limit=50                // Items per page. Default: 50. Max: 500.
?cursor=eyJpZCI6InVpZCJ9 // Opaque cursor from previous response Link header
?offset=100              // Offset-based (deprecated for cubes > 10K rows)
?order_by=updated_at    // Attribute key to sort by
?order_dir=desc         // asc | desc. Default: desc for updated_at, asc otherwise
?include_archived=false // Include soft-archived rows. Default: false.
?include_shadow=false   // Include ShadowRows from linked Grids. Default: false.

// Paginated response envelope
{
  "data": [...],          // Array of results
  "meta": {
```

```
    "total":      1842,      // Total matching rows (estimated for large cubes)
    "count":      50,        // Items in this page
    "cursor":      "eyJpZCI6InVpZCJ9", // cursor for next page (null on last
page)
    "next_cursor": "eyJpZCI6InVpZDIifQ", // explicit next cursor
    "has_more":    true,      // false on last page
    "limit":       50,
    "offset":      0
  },
  "links": {
    "self": "https://grid.host/api/v1/cubes/id/rows?limit=50",
    "next": "https://grid.host/api/v1/cubes/id/rows?limit=50&cursor=eyJ...",
    "prev": null,             // null on first page
    "first": "https://grid.host/api/v1/cubes/id/rows?limit=50",
    "last":  "https://grid.host/api/v1/cubes/id/rows?limit=50&cursor=last"
  }
}
```

4.4 Error Response Schema

```
// All errors follow this structure
{
  "error": {
    "code":      "not_found",          // Machine-readable error code
(snake_case)
    "message":    "Hypercube not found", // Human-readable message
    "detail":     "Cube ID abc-123 does not exist in grid xyz-456", // Additional
context
    "field":      "cube_id",           // Field that caused the error
(validation errors)
    "request_id": "req-abc-123",        // X-Request-ID echo
    "docs_url":   "https://docs.hypergrid.io/errors/not_found",
    "meta": {                          // Extra context (present on some
errors)
      "required_tier": 4,              // For insufficient_tier errors
      "current_tier":  2,
      "policy_id":     "pol-uuid",     // For policy_denied errors
      "conflict_id":   "conf-uuid",    // For version_conflict errors
      "retry_after":   30              // For rate_limit_exceeded errors
(seconds)
    }
  }
}
```

4.5 Complete Error Code Registry

HTT P	Code	Description
400	bad_request	Malformed request syntax, invalid JSON, or structurally invalid body.
400	validation_error	Request body or query params fail schema validation. See error.field.
400	invalid_dim_coordinate	The N-dim coordinate in the request doesn't match the Hypercube's declared dimensionality.

HTT P	Code	Description
400	invalid_hyperql	HyperQL query syntax error. See error.detail for parser error position.
400	invalid_crdt_operation	CrdtOperation payload is malformed or references an unknown attribute key.
400	cyclic_dependency	Adding this Hierarchy or Dependency edge would create a cycle in the graph.
400	schema_conflict	Proposed schema change conflicts with existing schema (e.g. duplicate axis name).
400	formula_invalid	Formula expression in AttributeKeyDef has a syntax or semantic error.
400	idempotency_conflict	An X-Idempotency-Key was reused with a different request body.
401	unauthorized	Missing, malformed, or unparseable auth token.
401	token_expired	JWT has expired. Refresh with POST /auth/token/refresh.
401	token_revoked	Token has been revoked via logout or key rotation.
403	insufficient_tier	Caller's PermissionTier is below the minimum required for this operation.
403	scope_not_granted	Required scope is absent from the token's scopes array.
403	cube_restricted	Token has cube_restrictions and this cube is not in the allowed list.
403	isolation_violation	Cross-partition access to HardIsolated partition attempted without authorization.
403	policy_denied	PolicyEngine evaluation blocked this operation. See meta.policy_id.
403	governance_required	This mutation requires a GovernanceProposal vote. See meta for proposal type.
403	consent_required	This edge type requires target consent before proceeding.
403	shadow_read_only	Write attempted on a ShadowRow attribute not in write_back_attrs.
404	not_found	Requested resource does not exist or is not visible to the caller.
404	cube_not_found	Hypercube ID does not exist in this Grid.
404	row_not_found	D _i key not found in this Hypercube.
404	cell_not_found	N-dim coordinate not found in the Hypercube (sparse cube, coordinate is empty).
404	edge_not_found	HypergraphEdge ID not found.
404	space_not_found	Space ID not found or caller is not a member.
404	namespace_not_found	NamespacePath does not resolve to any known entity.
404	snapshot_not_found	Snapshot ID not found or has expired from the snapshot store.
409	version_conflict	If-Match ETag does not match current entity version. See meta.conflict_id.
409	already_exists	Entity with this ID or namespace path already exists.
409	invalid_state_transition	Attempted lifecycle state transition is not valid in the configured Lattice.

HTTP	Code	Description
409	merge_conflict	CRDT merge produced a conflict. Write was recorded but conflict needs review.
422	unsupported_encoding	Requested StorageEncoding is incompatible with the declared axis types.
422	dim_limit_exceeded	Attempt to add more than 16 DimensionAxes to a Hypercube.
422	formula_resource_limit	Formula evaluation exceeded allowed cell refs, rows, depth, time, or memory.
429	rate_limit_exceeded	Too many requests. See X-RateLimit-* headers. Retry after meta.retry_after seconds.
500	internal_error	Unexpected server error. Idempotent requests may be safely retried.
502	upstream_error	Upstream dependency (PostgreSQL, Redis, Kafka) returned an error.
503	service_unavailable	Service temporarily unavailable (startup, maintenance). Retry-After header present.
504	gateway_timeout	Request timed out. For long-running HyperQL queries, use async query endpoint.

Part II — Grid API

5. Grids

A Grid is the root container for a complete Hypergrid deployment. It owns all Hypercubes, the Hypergraph, EventLog, CrdtLog, NamespaceRegistry, Spaces, and federation peer relationships. Each domain system (Kogi, Ume, Qala) is a separate Grid deployment.

5.1 Grid Object Schema

```
{
  "grid_id": "uuid",
  "name": "kogi-production",
  "version": "1.0.0", // Hypergrid substrate version
  "namespace_path": "hypergrid://kogi-production/",
  "domain_system": "kogi", // kogi | ume | qala | standalone
  "node_id": "node-us-east-1",
  "status": "active", // provisioning | active | maintenance |
  deprecated
  "cube_count": 42,
  "federation_peers": ["uuid-ume", "uuid-qala"],
  "space_count": 1204,
  "row_estimate": 84200000, // approximate total HyperRows across
  all cubes
  "created_at": "2026-01-15T09:00:00Z",
  "updated_at": "2026-03-22T14:30:00Z",
  "meta": {
    "description": "Kogi Independent Worker OS Grid",
    "region": "us-east-1",
    "contact_email": "ops@kogi.io"
  }
}
```

GET

/api/v1/grids

List all Grids accessible to the authenticated caller, including federated peer Grids.

Parameters:

Name	In	Type	Required	Description
limit	query	integer	No	Max results. Default: 20. Max: 100.
cursor	query	string	No	Pagination cursor from previous response.
include_peers	query	boolean	No	Include federated peer Grids. Default: false.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ data: Grid[], meta: PaginationMeta }
401	Unauthorized	{ error: { code: 'unauthorized', ... } }

GET	<code>/api/v1/grids/{grid_id}</code>
------------	--------------------------------------

Get full details of a specific Grid by ID.

Parameters:

Name	In	Type	Required	Description
grid_id	path	uuid	Yes	Grid identifier.

Responses:

Status	Meaning	Response Body
200 OK	Success	Grid object
404	Not found	{ error: { code: 'not_found', ... } }

POST	<code>/api/v1/grids</code>
-------------	----------------------------

Provision a new Grid deployment. Admin-tier operation.

Parameters:

Name	In	Type	Required	Description
X-Idempotency-Key	header	string	No	Idempotency key for safe retries.

Request Body:

```
{
  "name": "my-grid",           // required: URL-safe grid name
  "domain_system": "kogi",     // kogi | ume | qala | standalone
  "description": "My grid",
  "region": "us-east-1",      // deployment region
  "encoding_default": "Hybrid", // default StorageEncoding for new cubes
  "plugin_ids": ["time-axis", "geo-axis", "hierarchy-axis"],
  "meta": {}
}
```

Responses:

Status	Meaning	Response Body
201 Created	Grid provisioned	Grid object
400	Validation error	{ error: { code: 'validation_error', field: '...', ... } }
403	Insufficient tier	{ error: { code: 'insufficient_tier', ... } }
409	Already exists	{ error: { code: 'already_exists', ... } }
Note: Grid provisioning is an async operation. The Grid transitions from status =provisioning to status =active once boots trap completes. Poll GET /grids/{id} until status =active.		

GET	/api/v1/grids/{grid_id}/health
-----	--------------------------------

Get health and telemetry summary for a Grid: cell store connectivity, CRDT lag, federation peer status, AI engine status.

Parameters:

Name	In	Type	Required	Description
grid_id	path	uuid	Yes	Grid identifier.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ status, cell_store, crdt_lag_ms, peers: [{peer_id, status, lag_ms}], ai_engines: [{id, status}] }
503	Grid degraded	Health object with non-healthy status fields

Part III — Hypercube API

6. Hypercubes

A Hypercube is an N-dimensional grid: $H = (D_1, D_2, \dots, D_N)$. It is the generalization of a spreadsheet sheet. Hypercubes own a set of DimensionAxes and an AttributeKeyRegistry that defines the schema of every HyperCell within the cube. The actual cell data lives in the Grid's CellStoreBackend.

6.1 Hypercube Object Schema

```
{
  "cube_id":          "uuid",
  "grid_id":          "uuid",
  "name":             "kogi.portfolio.components",
  "namespace_path":   "hypergrid://kogi/personal/alice/cubes/components/",
  "space_id":         "uuid",           // owning Space (nullable)
  "n":                2,                // dimensionality (2..=16)
  "dims": [
    {
      "axis_id":       "uuid",
      "dim_index":     1,
      "name":          "entity",
      "axis_type":     "EntityAxis",      //
      "key_type":      "Uuid",
      "key_cardinality": "Dense",
      "ordering":      "Unordered",
      "nullable":      false,
      "index_strategy": "Hash",
      "plugin_id":     null
    },
    {
      "axis_id":       "uuid",
      "dim_index":     2,
      "name":          "property",
      "axis_type":     "PropertyAxis",
      "key_type":      "String",
      "key_cardinality": "Sparse",
      "ordering":      "Lexicographic",
      "nullable":      false,
      "index_strategy": "GIN",
      "plugin_id":     null
    }
  ],
  "encoding":         "Hybrid",          // Dense|Sparse|Hybrid
  "attr_count":       48,                // registered attribute keys
  "row_count":        2840000,           // estimated HyperRows (D1
  cardinality)
  "schema_version":   3,
  "status":           "active",          // active|archived|migrating
  "visibility":       "Tenant",
  "plugin_ids":       ["time-axis"],
}
```

```
"governance": {
  "require_vote_for": ["remove_attr", "change_visibility"]
},
"created_at":      "2026-01-15T09:00:00Z",
"updated_at":      "2026-03-22T14:30:00Z"
}
```

GET

/api/v1/grids/{grid_id}/cubes

List all Hypercubes in a Grid accessible to the caller.

Parameters:

Name	In	Type	Required	Description
grid_id	path	uuid	Yes	Grid identifier.
space_id	query	uuid	No	Filter to cubes owned by a specific Space.
name_prefix	query	string	No	Filter by cube name prefix (e.g. 'kogi.portfolio').
limit	query	integer	No	Max results. Default: 50. Max: 200.
cursor	query	string	No	Pagination cursor.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ data: Hypercube[], meta: PaginationMeta }
403	Forbidden	Caller has no access to this Grid.

Note: This endpoint returns cube schemas, not cell data. To query cells, use the Hyper Rows or Hyper QL

Status	Meaning	Response Body
endpoints.		

POST	/api/v1/grids/{grid_id}/cubes
------	-------------------------------

Create a new Hypercube in a Grid. Registers dimension axes and initial attribute keys.

Parameters:

Name	In	Type	Required	Description
grid_id	path	uuid	Yes	Grid identifier.
X-Idempotency-Key	header	string	No	Idempotency key.

Request Body:

```
{
  "name": "acme.crm.contacts", // required: unique within grid,
  "space_id": "uuid", // optional: owning Space
  "dims": [ // required: at least 2 dimensions
    {
      "dim_index": 1,
      "name": "contact",
      "axis_type": "EntityAxis", //
      "key_type": "Uuid", //
      "key_cardinality": "Dense", //
      "nullable": false, //
      "index_strategy": "Hash" //
    },
    {
      "dim_index": 2,
      "name": "property",
      "axis_type": "PropertyAxis",
      "key_type": "String",
      "key_cardinality": "Sparse",
      "nullable": false,
      "index_strategy": "GIN"
    }
  ],
  "encoding": "Hybrid", // Dense|Sparse|Hybrid
  "attr_keys": [ // optional: initial attribute keys
    {
      "key": "name",
      "display_name": "Contact Name",
      "attr_type": "Text",
      "write_permission": 3, // Contributor+
      "crdt_semantics": "LastWriteWins",
      "visibility": "Tenant"
    }
  ]
}
```



```
    }
  ],
  "plugin_ids": [],
  "visibility": "Tenant",
  "governance": { "require_vote_for": ["remove_attr"] }
}
```

Responses:

Status	Meaning	Response Body
201 Created	Cube created	Hypercube object
400	Validation error	Invalid dim config, duplicate axis index, unsupported axis_type.
403	Insufficient tier	Requires Editor+ tier.
409	Already exists	Cube with this name already exists in this Grid.
422	Unsupported encoding	Hybrid encoding not compatible with declared axis types.

Note:
Creating a Hypercube is a zero-downtime schema operation. The cube is immediately queryable once created. Add attribute keys via POST /cubes/{id}/

Status	Meaning	Response Body
attrs without any downtime.		

GET `/api/v1/cubes/{cube_id}`

Get complete schema for a Hypercube: all dimension axes and the full AttributeKeyRegistry.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.
include_attrs	query	boolean	No	Include full AttributeKeyRegistry in response. Default: true.

Responses:

Status	Meaning	Response Body
200 OK	Success	Hypercube object with dims[] and attr_registry[]
404	Not found	Cube does not exist or is not visible to caller.

PATCH `/api/v1/cubes/{cube_id}`

Update Hypercube metadata: name, description, visibility, governance config. Does not modify dims or attr_keys.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.

Request Body:

```
{
  "name": "acme.crm.contacts.v2", // optional: rename (creates ns alias)
  "description": "CRM contact records", // optional
  "visibility": "Public", // optional:
  Public|Tenant|Identity|Private
  "governance": { "require_vote_for": ["remove_attr", "change_visibility"] }
}
```

Responses:

Status	Meaning	Response Body
200 OK	Updated	Updated Hypercube object
403	Insufficient tier	Requires Manager+ tier.
409	Governance required	Visibility change requires GovernanceProposal vote.

DELETE	<code>/api/v1/cubes/{cube_id}</code>
---------------	--------------------------------------

Archive a Hypercube. Sets `status=archived`. Does not delete data — all cells remain accessible via `AS_OF` queries and the `EventLog`.

Parameters:

Name	In	Type	Required	Description
<code>cube_id</code>	path	uuid	Yes	Hypercube identifier.

Responses:

Status	Meaning	Response Body
204 No Content	Archived	No body.
403	Insufficient tier	Requires Owner tier.
409	Governance required	Archiving this cube requires a GovernanceProposal vote.

Note: Archiving is irreversible without an Owner restoring the cube via PATCH status=`active`. All data

Status	Meaning	Response Body
is preserved.		

6.2 Dimension Axis API

GET	/api/v1/cubes/{cube_id}/dims
-----	------------------------------

List all *DimensionAxes* for a *Hypercube*.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ data: DimensionAxis[] }

POST	/api/v1/cubes/{cube_id}/dims
------	------------------------------

Add a new *DimensionAxis* to an existing *Hypercube* ($N+1$). Zero-downtime schema evolution. Existing cells gain the new dimension with null/default key.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.
X-Idempotency-Key	header	string	No	Idempotency key.

Request Body:

```
{
  "dim_index":      3,                      // must be current_n + 1
  "name":           "time_period",
  "axis_type":      "TimeAxis",
  "key_type":       "Timestamp",
  "key_cardinality": "Sparse",
  "nullable":       true,
  "ordering":       "Chronological",
  "index_strategy": "BRIN",                // BRIN recommended for TimeAxis
  "plugin_id":      "time-axis"           // built-in time-axis plugin
}
```

Responses:

Status	Meaning	Response Body
201 Created	Axis added	DimensionAxis object
400	Invalid dim_index	dim_index must equal current n+1.
403	Insufficient tier	Requires Manager+ tier.
409	Governance required	Adding a dimension may require a GovernanceProposal vote.
422	Dim limit exceeded	Hypercube already has 16 dimensions (maximum).
Note: Adding a dimension is a SchemaEvent written to the SchemaEventLog. All federation peers receive and apply the AddDimension SchemaEvent automatically.		

GET	/api/v1/cubes/{cube_id}/dims/{axis_id}
-----	--

Get a specific DimensionAxis by ID.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.
axis_id	path	uuid	Yes	DimensionAxis identifier.

Responses:

Status	Meaning	Response Body
200 OK	Success	DimensionAxis object
404	Not found	Axis does not exist on this cube.

PATCH `/api/v1/cubes/{cube_id}/dims/{axis_id}`

Update a DimensionAxis: rename, change index strategy, update ordering. Does not change axis_type or key_type (breaking — requires migration).

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.
axis_id	path	uuid	Yes	DimensionAxis identifier.

Request Body:

```
{
  "name": "fiscal_quarter", // rename (creates alias)
  "index_strategy": "BRIN", // update index strategy
  "description": "Fiscal quarter key"
}
```

Responses:

Status	Meaning	Response Body
200 OK	Updated	Updated DimensionAxis
400	Validation error	Cannot change axis_type or key_type via PATCH.
403	Insufficient tier	Requires Manager+ tier.

6.3 Attribute Key Registry API

GET `/api/v1/cubes/{cube_id}/attrs`

List all registered AttributeKeyDefs for a Hypercube.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.
attr_type	query	string	No	Filter by AttributeType (e.g. AI, Computed, Text).
dimension_scope	query	string	No	Filter by dimension scope (e.g. axis_id).

Responses:

Status	Meaning	Response Body
200 OK	Success	{ data: AttributeKeyDef[] }

POST `/api/v1/cubes/{cube_id}/attrs`

Register a new attribute key in the Hypercube's AttributeKeyRegistry. Zero-downtime. Existing cells return default_value for the new key.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.
X-Idempotency-Key	header	string	No	Idempotency key.

Request Body:

```
{
  "key": "lead_score", // required: unique within cube
  "display_name": "Lead Score",
  "attr_type": "AI", //
  Text|Number|Currency|Percent|Bool|Date| //
  // DateTime|Duration|Enum|MultiEnum|
  // Relation|MultiRelation|User|Tag|
  //
  Json|Computed|AI|Formula|Audit|Custom
  "description": "AI-computed lead quality score (0-100)",
  "dimension_scope": "AllDimensions", // AllDimensions | { specific_axes:
  [axis_id] }
  "default_value": null,
  "required": false,
  "visibility": "Tenant", // Public|Tenant|Identity|Owner
  "write_permission": 7, // System (AI-write only)
  "crdt_semantics": "LastWriteWins", //
  LWW|OR-Set|GrowOnlyCounter|PNCOUNTER| //
  // MaxRegister|MinRegister|Lattice|
  // AppendOnlyLog|JsonCrdt|Custom
  // for Computed and AI attr_types
  "computation": {
    "type": "AiEngine",
    "engine_id": "crm.lead_score_engine.v1",
    "model_name": "lead_score_v2",
  }
}
```

```
    "input_attrs":
    ["stage","last_contact_at","value","source","interaction_count"],
    "staleness_ttl": 7200           // seconds before AI signal considered
    stale
  },
  "aggregation":      ["AVG","MIN","MAX"],
  "index_strategy":   "BTree",
  "searchable":       false,
  "version_track":    true,
  "export_label":     "Lead Quality Score"
}
```

Responses:

Status	Meaning	Response Body
201 Created	Attr key registered	AttributeKeyDef object
400	Validation error	Invalid crdt_semantics for attr_type, duplicate key name, etc.
403	Insufficient tier	Requires Editor+ tier.
409	Already exists	Attribute key with this name already registered.

Note: Registering an attribute key is a SchemaEvent. All federation peers receive and apply it automatically. Existing cells get default_value for this key

Status	Meaning	Response Body
on next read.		

GET	<code>/api/v1/cubes/{cube_id}/attrs/{attr_key}</code>
------------	---

Get a specific *AttributeKeyDef* by key name.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.
attr_key	path	string	Yes	Attribute key name (e.g. 'health_score').

Responses:

Status	Meaning	Response Body
200 OK	Success	AttributeKeyDef object
404	Not found	Attr key not registered on this cube.

PATCH	<code>/api/v1/cubes/{cube_id}/attrs/{attr_key}</code>
--------------	---

Update an *AttributeKeyDef*: rename (with alias), change *display_name*, update *default_value*, change *write_permission*, update aggregation list. Cannot change *crdt_semantics* of an existing key without schema version bump.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.
attr_key	path	string	Yes	Attribute key name.

Request Body:

```
{
  "display_name": "Lead Quality Score",
  "description": "Updated description",
  "default_value": 0.0,
  "visibility": "Tenant",
  "write_permission": 7,
  "export_label": "Score"
}
```

Responses:

Status	Meaning	Response Body
200 OK	Updated	Updated AttributeKeyDef
400	Validation error	crdt_semantics change not allowed via PATCH.
403	Insufficient tier	Requires Manager+ tier.

DELETE	<code>/api/v1/cubes/{cube_id}/attrs/{attr_key}</code>
---------------	---

Tombstone (soft-delete) an attribute key. The key is marked as tombstoned and removed from schema validation. Existing cell values are preserved in the EventLog for AS_OF queries but not returned in live queries.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.
attr_key	path	string	Yes	Attribute key name.

Responses:

Status	Meaning	Response Body
204 No Content	Tombstoned	No body.
403	Insufficient tier	Requires Owner tier.
409	Governance required	Removing an attribute key requires GovernanceProposal vote if cube.governance.require_vote_for includes 'remove_attr'.

Note: Tombstoning is irreversible. Historical values for this key remain accessible via AS_OF

Status	Meaning	Response Body
queries. Active cells will no longer return this key.		

Part IV — HyperRow and HyperCell API

7. HyperRows

A HyperRow is all cells sharing the same D₁ key — the canonical entity in a Hypercube. It is the generalization of a spreadsheet row. The HyperRow API provides entity-level CRUD with CRDT-safe writes, pagination, filtering, and time-travel.

7.1 HyperRow Object Schema

```
{
  "d1_key":      "uuid",                                // D1 key - entity identifier
  "cube_id":     "uuid",
  "grid_id":     "uuid",
  "attributes": {                                       // attribute map: key → TypedAttrValue
    "name":      { "value": "Acme Corp", "type": "Text",
                  "version": 14, "last_actor": "user-1:alice:node-1",
                  "updated_at": "2026-03-22T10:00:00Z" },
    "status":    { "value": "Active", "type": "Enum",
                  "version": 3, "last_actor": "user-1:alice:node-1",
                  "updated_at": "2026-03-20T08:00:00Z" },
    "tags":      { "value": ["enterprise","tier-1"], "type": "TagSet",
                  "version": 7, "crdt": "OR-Set" },
    "lead_score": { "value": 87.4, "type": "AI",
                  "computed_by": "crm.lead_score_engine.v1",
                  "computed_at": "2026-03-22T09:50:00Z",
                  "confidence": 0.92 },
    "budget":    { "value": { "amount": "50000.00", "currency": "USD" },
                  "type": "Currency" }
  },
  "vector_clock": { "node-us-east-1": 42, "node-eu-west-1": 37 },
  "version":      55,
  "visibility":   "Tenant",
  "created_at":   "2026-01-20T09:00:00Z",
  "updated_at":   "2026-03-22T10:00:00Z",
  "is_shadow":    false,
  "edge_refs":    ["edge-uuid-1", "edge-uuid-2"]
}
```

GET

/api/v1/cubes/{cube_id}/rows

List HyperRows in a Hypercube with optional DimSlice filtering, sorting, grouping. Returns paginated results.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.

Name	In	Type	Required	Description
filter	query	string	No	DimSlice predicate in HyperQL WHERE syntax. URL-encoded. Example: D2='status' AND value='Active'
sort	query	string	No	Sort: {attr_key}:{asc desc}. Example: updated_at:desc
group_by	query	string	No	Group by attr_key. Returns grouped result structure.
attrs	query	string	No	Comma-separated list of attr_keys to include. Default: all. Example: name,status,health_score
include_shadow	query	boolean	No	Include ShadowRows from linked Grids. Default: false.
identity_tag	query	string	No	Scope to a specific identity partition. Example: @alice-dev
space_id	query	uuid	No	Scope to rows tagged to a specific Space.
as_of	query	datetime	No	Time-travel: return row state at this UTC timestamp. ISO 8601.
limit	query	integer	No	Page size. Default: 50. Max: 500.
cursor	query	string	No	Pagination cursor.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ data: HyperRow[], meta: PaginationMeta }
400	Invalid filter	Malformed DimSlice predicate syntax. See error.detail.
404	Cube not found	

Note:
For complex queries spanning multiple cubes, use the HyperQL endpoint POST /cubes/{id}/

Status	Meaning	Response Body
query instead.		

POST	/api/v1/cubes/{cube_id}/rows
------	------------------------------

Create a new HyperRow. Accepts a D1 key and initial attribute values. Produces SetAttr CrdtOperations for all provided attributes.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.
X-Idempotency-Key	header	string	No	Idempotency key. Recommended to prevent duplicate creation.

Request Body:

```
{
  "d1_key": "uuid", // optional: server-generated if omitted
  "attributes": {
    "name": "Acme Corp",
    "status": "Active",
    "tags": ["enterprise","tier-1"], // OR-Set: each tag added as separate
OR-Set Add
    "budget": { "amount": "50000.00", "currency": "USD" },
    "owner": "user-uuid"
  },
  "visibility": "Tenant", // row-level visibility
  "space_id": "uuid", // optional Space association
  "identity_tags": ["@alice-dev"] // optional: partition tags
}
```

Responses:

Status	Meaning	Response Body
201 Created	Row created	HyperRow object with server-assigned d1_key
400	Validation error	Required attributes missing, type mismatch.
403	Insufficient tier	Requires Contributor+ tier.
409	Already exists	A row with this d1_key already exists.
Note: All attribute writes produce		

Status	Meaning	Response Body
ce CrdtO perati ons and are recor ded in the Event Log. Set-v alued attrib utes (TagS et, Multi Enum , Multi Relati on) are initiali zed with AddT oSet OR-S et opera tions, one per elem ent.		

GET	/api/v1/cubes/{cube_id}/rows/{d1_key}
-----	---------------------------------------

Get all cells for a single HyperRow by D₁ key. Returns the complete attribute map.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.
d1_key	path	uuid	Yes	D ₁ key (entity identifier).
attrs	query	string	No	Comma-separated attr keys to include.

Name	In	Type	Required	Description
as_of	query	datetime	No	Time-travel: return state at this timestamp.
include_computed	query	boolean	No	Include Tier-1 formula-computed attrs. Default: true.
include_ai	query	boolean	No	Include Tier-2 AI-computed attrs. Default: true.

Responses:

Status	Meaning	Response Body
200 OK	Success	HyperRow object
404	Not found	DI key not found in this cube.

Note:
For as_of queries, the server performs Event Log replay from the nearest snapshots hot. For hot cubes, this may take up to 500ms. Use async queries for historical analysis

Status	Meaning	Response Body
workloads.		

PATCH	/api/v1/cubes/{cube_id}/rows/{d1_key}
-------	---------------------------------------

Update attributes of a HyperRow. Each attribute write generates the appropriate CrdtOperation (SetAttr, AddToSet, IncrCounter, TransitionState). Partial update — only listed attributes are changed.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.
d1_key	path	uuid	Yes	D1 key.
If-Match	header	string	No	ETag / VectorClock version for optimistic concurrency.

Request Body:

```
{
  "attributes": {
    "name": "Acme Corporation", // SetAttr LWW
    "status": "Inactive", // TransitionState (if Lattice CRDT)
    "tags": {
      "add": ["fortune-500"], // AddToSet OR-Set
      "remove": ["tier-1"] // RemoveFromSet OR-Set (by value lookup)
    },
    "restart_count": { "incr": 1 }, // IncrCounter (GrowOnly)
    "budget": { "amount": "75000.00", "currency": "USD" }
  }
}
```

Responses:

Status	Meaning	Response Body
200 OK	Updated	Updated HyperRow object with new vector_clock
400	Invalid state transition	Attempted lifecycle transition not valid in Lattice.
403	Insufficient tier	Requires Editor+ tier for most attrs. Specific attrs may require higher tier.
403	Shadow read-only	Attempted write to shadow attr not in write_back_attrs.
409	Version conflict	If-Match ETag does not match current version.

Note: CRDT

Status	Meaning	Response Body
conflicts (concurrent writes from multiple nodes) do NOT return 409. They are resolved per CRD semantics and a ConflictRecord is generated. The X-Conflict-ID response header will be set if a conflict was recorded.		

DELETE	/api/v1/cubes/{cube_id}/rows/{dl_key}
--------	---------------------------------------

Soft-archive a HyperRow. Sets status=Archived. The row remains accessible via as_of queries and the EventLog. Does not delete the physical cell data.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.
d1_key	path	uuid	Yes	D1 key.

Responses:

Status	Meaning	Response Body
204 No Content	Archived	No body.
403	Insufficient tier	Requires Owner tier.

Note: Physical deletion is not supported in Hypergrid. All data is preserved in the Event Log for audit and time-travel. To hard-delete (GDPR erasure), use the GDPR erasure endpoint

Status	Meaning	Response Body
POST /identity/erase.		

8. HyperCells — N-Dimensional Coordinate API

The HyperCell API provides direct access to individual N-dimensional coordinates. Use this when you need to read or write a specific cell at a full coordinate tuple, including higher dimensions (D₃, D₄...). For entity-level operations, prefer the HyperRow API.

GET	/api/v1/cubes/{cube_id}/cells/{coord}
-----	---------------------------------------

Get a single HyperCell at a specific N-dimensional coordinate. The coord path parameter is a comma-separated tuple of dimension key values in order D₁..D_N.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.
coord	path	string	Yes	Comma-separated coordinate tuple. Example: 'uuid-entity,health_score,2026-Q2,EMEA' for N=4.
as_of	query	datetime	No	Time-travel: return cell state at this timestamp.

Responses:

Status	Meaning	Response Body
200 OK	Success	HyperCell object with attributes map
404	Not found	No cell exists at this coordinate (sparse cube — coordinate is empty). Not an error for writes — use PUT to create.

Note:
For sparse cubes, a 404 at a coordinate means no

Status	Meaning	Response Body
	data exists there. This is normal for Time Axis and GeoAxis dimensions where most coordinate combinations are empty.	

PUT	/api/v1/cubes/{cube_id}/cells/{coord}
-----	---------------------------------------

Write a complete attribute map to a specific N-dimensional coordinate. Creates the cell if it doesn't exist. All provided attributes are written as SetAttr CrdtOperations.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.
coord	path	string	Yes	Comma-separated N-dim coordinate tuple.

Request Body:

```
{
  "attributes": {
    "value": 87.4,
    "data_type": "Number",
    "format": "0.0",
    "computed_by": "kogi.health_score_engine.v2",
    "confidence": 0.92
  }
}
```

Responses:

Status	Meaning	Response Body
200 OK	Updated/Created	HyperCell object
400	Invalid coordinate	Coordinate tuple length does not match cube.n.
403	Insufficient tier	Requires Editor+ or System tier for AI attrs.

PATCH `/api/v1/cubes/{cube_id}/cells/{coord}/{attr_key}`

Update a single attribute at a specific N-dimensional coordinate. Generates the appropriate CrdtOperation based on the attr_key's configured CrdtSemantics.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.
coord	path	string	Yes	N-dim coordinate tuple.
attr_key	path	string	Yes	Attribute key to update.

Request Body:

```
// For LWW attrs:
{ "value": "new_value" }

// For OR-Set attrs (TagSet, MultiEnum, MultiRelation):
{ "add": ["tag1","tag2"], "remove": ["old-tag"] }

// For Counter attrs:
{ "incr": 5 }

// For Lattice (status transition):
{ "from": "Active", "to": "Deprecated" }

// For JSON CRDT:
{ "deep_merge": { "config_key": "new_value" } }
```

Responses:

Status	Meaning	Response Body
200 OK	Updated	{ cell: HyperCell, conflict_id: null uuid }
400	Invalid operation	incr < 0 on GrowOnly counter, invalid state transition, etc.
403	Permission denied	write_permission tier not met for this attr_key.

Note:
If the
write
produ
ces a

Status	Meaning	Response Body
CRDT conflict (concurrent write from another node arrived before this one), the response still returns 200 OK but includes a non-null conflict_id in the response body. The X-Conflict-ID response header is also set.		

GET	/api/v1/cubes/{cube_id}/rows/{dl_key}/history
-----	---

Get the full EventLog for a HyperRow: all mutations, actors, timestamps, before/after values. Paginated, most recent first.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.
d1_key	path	uuid	Yes	D1 key.
attr_key	query	string	No	Filter to changes of a specific attribute key.
since	query	datetime	No	Only return events after this timestamp.
until	query	datetime	No	Only return events before this timestamp.
limit	query	integer	No	Default: 100. Max: 1000.
cursor	query	string	No	Pagination cursor.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ data: EventLogEntry[], meta: PaginationMeta }
403	Insufficient tier	Requires Editor+ tier.
404	Not found	HyperRow not found.

Note: Event Log entries are immutable and cannot be deleted. For GDPR erasure requests, the actor_id in Event Log entries is anonymized but the struct

Status	Meaning	Response Body
ural recor d (what chan ged, when) is prese rved.		

Part V — HyperQL Query API

9. HyperQL — N-Dimensional Query Language

HyperQL is Hypergrid's native N-dimensional query language. It extends SQL-like syntax with operations specific to N-dimensional grids: DimSlice filtering, DimFold aggregation, DimExpand pivot, TRAVERSE GRAPH Hypergraph traversal, AS_OF time-travel, and INCLUDE SHADOW CELLS for cross-Grid queries. The HyperQL API exposes synchronous (small result sets) and asynchronous (large analytical queries) execution modes.

9.1 Synchronous Query

POST	/api/v1/cubes/{cube_id}/query
------	-------------------------------

Execute a HyperQL query synchronously against a specific Hypercube. For result sets under 100,000 rows. Larger queries should use the async endpoint.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Target Hypercube. Can be overridden in the HyperQL query FROM clause.
timeout_ms	query	integer	No	Max query execution time. Default: 10,000ms. Max: 30,000ms.

Request Body:

```
{
  "query": "SELECT D1.entity_id, cell[D1,'name'].value AS name,
cell[D1,'health_score'].value AS score FROM kogi.portfolio.components WHERE
cell[D1,'status'].value = 'Active' AND cell[D1,'health_score'].value > 70 ORDER
BY cell[D1,'health_score'].value DESC LIMIT 20",

  // Optional: explicitly set query options (override inline)
  "options": {
    "as_of": "2026-03-01T00:00:00Z", // time-travel
    "identity_tag": "@alice-dev", // partition scope
    "space_id": "uuid", // Space scope
    "include_shadow": false, // include shadow cells?
    "max_rows": 10000 // row limit guard
  }
}
```

Responses:

Status	Meaning	Response Body
200 OK	Query executed	{ columns: [{ name, type }], rows: [...], meta: { rows_returned, execution_ms, plan_notes } }
400	HyperQL parse error	{ error: { code: 'invalid_hyperql', detail: 'Unexpected token at position 42', ... } }
400	Validation error	Invalid as_of timestamp, unknown cube name, unregistered attr_key.
403	Insufficient scope	Caller does not have read access to referenced cubes.
504	Query timeout	Query exceeded timeout_ms. Use async endpoint for long-running queries.

Note: Result rows are arrays in column order matching the columns array. Null values are represented as JSON null. TypedAttr Value types are mapped to JSON types: Text → string, Number → number, Currency → {amo

Status	Meaning	Response Body
unt,c urren cy}, TagS et→st ring[], Json →obj ect.		

9.2 Asynchronous Query

POST	/api/v1/query/async
------	---------------------

Submit a HyperQL query for asynchronous execution. Returns a query_id immediately. Results can be polled or streamed via SSE. Use for large analytical queries, DimFold aggregations over millions of rows, or cross-cube joins.

Request Body:

```
{
  "query":      "SELECT D1.entity_id, FOLD D3 WITH SUM(value) AS total_revenue
FROM kogi.portfolio.finances WHERE D2.metric_name = 'revenue' AND D3.period
BETWEEN '2025-Q1' AND '2026-Q4' GROUP BY D1.entity_id ORDER BY total_revenue
DESC",
  "output_format": "json",                // json | ndjson | parquet | arrow | csv
  "callback_url": "https://app.example.com/webhooks/query", // optional webhook
  "options": {
    "max_rows": 1000000,
    "timeout_ms": 300000                // 5 minute timeout for async
  }
}
```

Responses:

Status	Meaning	Response Body
202 Accept ed	Query queued	{ query_id: 'uuid', status: 'queued', estimated_ms: 5000, poll_url: '/api/v1/query/async/{query_id}', stream_url: '/api/v1/query/async/{query_id}/stream' }
400	HyperQL parse error	
403	Insufficient scope	
Note: Asyn c queri es are execu		

Status	Meaning	Response Body
ted in a separate worker pool and do not block the request thread. Maximum result set: 10M rows. For larger datasets, use the export endpoint.		

GET	/api/v1/query/async/{query_id}
-----	--------------------------------

Poll status of an async query. Returns results inline when complete for result sets under 1MB. For larger results, use the stream endpoint.

Parameters:

Name	In	Type	Required	Description
query_id	path	uuid	Yes	Async query ID from POST /query/async.

Responses:

Status	Meaning	Response Body
200 OK	Pending	{ query_id, status: 'running', progress_pct: 42, started_at, estimated_ms_remaining }
200 OK	Complete (small result)	{ query_id, status: 'complete', columns, rows, meta: { rows_returned, execution_ms } }

Status	Meaning	Response Body
200 OK	Complete (large result)	{ query_id, status: 'complete', result_url: '/api/v1/query/async/{id}/download', rows_total, size_bytes }
200 OK	Failed	{ query_id, status: 'failed', error: { code, message, detail } }
404	Not found	Query ID expired (TTL: 1 hour) or not found.

GET	/api/v1/query/async/{query_id}/stream
-----	---------------------------------------

Stream results of a completed async query as NDJSON (newline-delimited JSON). Each line is one result row.

Parameters:

Name	In	Type	Required	Description
query_id	path	uuid	Yes	Completed async query ID.
format	query	string	No	ndjson (default) csv parquet

Responses:

Status	Meaning	Response Body
200 OK	Streaming	Content-Type: application/x-ndjson. Stream of row objects.
202 Accepted	Still running	Query not yet complete.
404	Not found	Query ID expired or not found.

Note: Set Accept: application/x-ndjson and process the response as a stream. Each line is a JSON

Status	Meaning	Response Body
object with column-name keys. The final line is a metadata summary: { '_meta': { 'rows_total', 'execution_ms', 'truncated' } }.		

9.3 Natural Language Query

POST	/api/v1/query/nl
------	------------------

Translate a natural language question into HyperQL and execute it. Powered by the registered AEngineAdapter NL-to-HyperQL bridge. Returns both the generated HyperQL and the query results.

Request Body:

```
{
  "question": "Which of my active projects are over budget this quarter?",
  "cube_context":["kogi.portfolio.components"], // which cubes to consider
  "identity_tag":"@alice-dev",
  "space_id": "uuid",
  "return_query":true // also return the generated HyperQL
}
```

Responses:

Status	Meaning	Response Body
200 OK	Success	{ generated_hyperql: string, columns, rows, meta: { confidence, execution_ms } }

Status	Meaning	Response Body
400	AI engine not configured	NL query requires a registered AIEngineAdapter.
400	Query translation failed	AI engine could not translate the question to HyperQL.

Note:
The generated HyperQL is returned in generated_hyperql for transparency. Users can copy it to the /cubes/{id}/query endpoint to execute it directly with modifications. Confidence score reflects AI certainty in the translation.

9.4 DimFold and DimExpand

POST	/api/v1/cubes/{cube_id}/fold
------	------------------------------

Execute a DimFold aggregation: collapse one dimension axis by aggregating all its key values into a scalar summary. The result is an (N-1)-dimensional projection.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.

Request Body:

```
{
  "fold_axis": "time_axis_id",           // axis to collapse
  "agg_fn": "SUM",                       //
SUM|AVG|MIN|MAX|COUNT|COUNT_UNIQUE|
                                           //
MEDIAN|PERCENTILE|STDDEV|DISTRIBUTION
  "attr_key": "value",
  "filter": {
    "d2_key": "revenue",                 // pre-fold filter on D2
    "d4_key_in": ["EMEA", "APAC"]       // pre-fold filter on D4
  },
  "group_by": ["d1_key"],                // optional post-fold grouping
  "order_by": "folded_value",
  "order_dir": "desc",
  "limit": 100
}
```

Responses:

Status	Meaning	Response Body
200 OK	Success	{ columns: [{ name, type }], rows: [...] } }
400	Invalid axis	Specified axis does not exist on this cube.

Note: DimFold is equivalent to FOLD D_n WITH AGG(attr_key) in HyperQL. For cubes with N>4

Status	Meaning	Response Body
or result sets over 1M rows, this opera tion is auto matic ally route d to the Click Hous e analyt ics backe nd.		

POST	/api/v1/cubes/{cube_id}/expand
------	--------------------------------

Execute a DimExpand pivot: expand one dimension's key set into separate result columns. Equivalent to a PIVOT or EXPAND D AS COLUMNS(...) in HyperQL.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.

Request Body:

```
{
  "expand_axis": "category_axis_id",           // axis to expand
  "value_expr": { "agg_fn": "AVG", "attr_key": "value" },
  "filter": {
    "d2_key": "score"
  },
  "group_by": ["d1_key", "d3_key"],             // remaining axes to group by
  "col_prefix": "cat_",                        // prefix for generated column names
  "limit": 500
}
```

Responses:

Status	Meaning	Response Body
200 OK	Success	{ columns: [{ name, type }], rows: [...] }

Status	Meaning	Response Body
Note: DimE xpan d dyna micall y gener ates one outpu t colum n per uniqu e key value in the expa nd_a xis. Colu mn name s are prefix ed with col_p refix if provi ded. For axes with high cardi nality (>100 uniqu e value s), set a filter first.		

Part VI — Hypergraph API

10. Hypergraph Nodes

Every entity in every Hypercube is automatically registered as a HypergraphNode. Nodes are created implicitly when HyperRows are created. The Nodes API provides direct access to node attributes and metadata.

GET	/api/v1/graph/nodes/{node_id}
-----	-------------------------------

Get a HypergraphNode by ID.

Parameters:

Name	In	Type	Required	Description
node_id	path	uuid	Yes	HypergraphNode identifier.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ node_id, node_type, entity_ref, attributes, namespace_path, visibility }
404	Not found	Node does not exist or is not visible to caller.

GET	/api/v1/graph/nodes/{node_id}/tree
-----	------------------------------------

Get the link tree rooted at a node: a directed subgraph of all reachable nodes up to max_depth via specified edge types. Respects VisibilityMask — hidden nodes silently excluded.

Parameters:

Name	In	Type	Required	Description
node_id	path	uuid	Yes	Root node identifier.
max_depth	query	integer	No	Maximum traversal depth. Default: 3. Max: 10.
min_depth	query	integer	No	Minimum depth to include. Default: 0.
edge_type	query	string	No	Filter to specific edge type. Comma-separated for multiple. Default: all types.
direction	query	string	No	Outbound Inbound Both. Default: Outbound.
include_attrs	query	boolean	No	Include node attributes in response. Default: false (IDs only for performance).

Name	In	Type	Required	Description
space_id	query	uuid	No	Filter to edges scoped to a specific Space.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ root: NodeId, nodes: [{ node_id, depth, node_type, entity_ref, attrs? }], edges: [{ edge_id, from, to, edge_type, weight }], total_nodes, total_edges }
404	Not found	Root node not found.
422	Depth limit	max_depth > 10 is not allowed.

Note:
For link forests (all trees rooted at an identity), use GET /graph/nodes/{id}/forest.
For the shortest path between two nodes, use GET /graph/path.

GET	/api/v1/graph/nodes/{node_id}/forest
-----	--------------------------------------

Get the complete link forest for an identity: all link trees rooted at all nodes owned by this identity. Returns the union of all their ego graphs.

Parameters:

Name	In	Type	Required	Description
node_id	path	uuid	Yes	Identity node identifier.
max_depth	query	integer	No	Tree depth. Default: 3.
edge_type	query	string	No	Edge type filter.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ identity_node_id, roots: NodeId[], nodes: [...], edges: [...], total_nodes, total_edges }
403	Insufficient tier	Requires Owner tier to see full forest.

GET	<code>/api/v1/graph/nodes/{node_id}/ego</code>
------------	--

Get the ego network (depth 1+2) centered on a node: all direct connections and their connections.

Parameters:

Name	In	Type	Required	Description
node_id	path	uuid	Yes	Center node identifier.
edge_type	query	string	No	Edge type filter.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ center: NodeId, depth_1: [...], depth_2: [...], edges: [...] }

11. Hypergraph Edges

11.1 Edge Object Schema

```
{
  "edge_id":      "uuid",
  "from_node":    "uuid",
  "to_node":      "uuid",
  "edge_type":    "Hierarchy", //
Hierarchy|Dependency|Association|Contains| // Derives|CrossGridLink|ShadowOf|
//
SpaceMembership|Collaborates|References|
//
FederationPeer|NamespaceAlias|ComputedFrom|
```

```

//
SubscribesTo|InvestedIn|Employs|Custom(String)
  "direction":      "Directed",           // Directed|Bidirectional
  "weight":         1.0,                  // 0.0-1.0
  "attributes":     {
    "role":         "contributor",        // edge-level attributes
    "since":        "2026-01-15T09:00:00Z"
  },
  "visibility":     "Tenant",
  "consent_status": "Accepted",           //
None|Pending|Accepted|Declined|Revoked
  "consent_config": {
    "mirrored_attrs": ["name","status","health_score"],
    "write_back_attrs": [],
    "update_policy":  "RealTime"          // RealTime|Batched|Manual
  },
  "space_scope":    null,                  // null = global
  "shadow_cell_id": "uuid",               // null if no shadow cell
  "expires_at":     null,
  "created_at":     "2026-01-20T09:00:00Z",
  "created_by":     "user-uuid:@alice:node-1"
}
```

GET **/api/v1/graph/edges**

List HypergraphEdges with optional filters.

Parameters:

Name	In	Type	Required	Description
from_node	query	uuid	No	Filter by source node.
to_node	query	uuid	No	Filter by target node.
edge_type	query	string	No	Filter by edge type. Comma-separated.
consent_status	query	string	No	Filter by consent status: Pending Accepted Declined Revoked.
space_id	query	uuid	No	Filter to edges scoped to a Space.
limit	query	integer	No	Default: 50. Max: 500.
cursor	query	string	No	Pagination cursor.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ data: HypergraphEdge[], meta: PaginationMeta }

POST **/api/v1/graph/edges**

Create a *HypergraphEdge*. For edge types that require consent (*CrossGridLink*, *Collaborates*, *InvestedIn*, *Employs*), the edge is created with *consent_status=Pending* and a consent invitation is sent to the target node owner.

Parameters:

Name	In	Type	Required	Description
X-Idempotency-Key	header	string	No	Idempotency key.

Request Body:

```
{
  "from_node": "uuid",
  "to_node": "uuid",
  "edge_type": "CrossGridLink",           // see EdgeType enum
  "direction": "Directed",
  "weight": 1.0,
  "attributes": { "relationship": "contractor" },
  "space_scope": null,
  "expires_at": null,

  // For CrossGridLink, Collaborates, InvestedIn, Employs:
  "consent_request": {
    "message": "I'd like to link our projects to share status updates.",
    "proposed_mirrored_attrs": ["name","status","health_score"],
    "proposed_write_back_attrs": [],
    "proposed_update_policy": "RealTime"
  }
}
```

Responses:

Status	Meaning	Response Body
201 Created	Edge created / consent pending	HypergraphEdge object with <i>consent_status=None</i> or <i>Pending</i>
400	Cyclic dependency	Adding this Hierarchy or Dependency edge would create a cycle.
403	Insufficient tier	Requires Contributor+ tier.
409	Already exists	An edge of this type between these nodes already exists.

Note: For edges requiring consent, the response has

Status	Meaning	Response Body
consent_status=Pending. The target node owner receives a consent invitation. Poll GET /graph/edges/{id} for status changes, or subscribe to the WebSocket space activity stream.		

GET	/api/v1/graph/edges/{edge_id}
-----	-------------------------------

Get a specific HypergraphEdge by ID.

Parameters:

Name	In	Type	Required	Description
edge_id	path	uuid	Yes	Edge identifier.

Responses:

Status	Meaning	Response Body
200 OK	Success	HypergraphEdge object
404	Not found	Edge does not exist or is not visible.

PATCH `/api/v1/graph/edges/{edge_id}`

Update edge attributes: *weight*, *attribute map*, *consent_config*, *expiry*. Cannot change *from_node*, *to_node*, or *edge_type*.

Parameters:

Name	In	Type	Required	Description
edge_id	path	uuid	Yes	Edge identifier.

Request Body:

```
{
  "weight":      0.8,
  "attributes":  { "role": "lead_contributor" },
  "consent_config": {
    "mirrored_attrs": ["name", "status", "health_score", "risk_score"],
    "update_policy":  "Batched"                // RealTime|Batched|Manual
  },
  "expires_at":  "2027-01-01T00:00:00Z"
}
```

Responses:

Status	Meaning	Response Body
200 OK	Updated	Updated HypergraphEdge
403	Insufficient tier	Requires Owner tier of this edge.

POST `/api/v1/graph/edges/{edge_id}/consent`

Accept or decline a pending consent invitation for a *CrossGridLink* or similar edge type. Called by the target node owner.

Parameters:

Name	In	Type	Required	Description
edge_id	path	uuid	Yes	Edge identifier.

Request Body:

```
{
  "decision":    "accept",                // accept | decline
}
```

```
"mirrored_attrs": ["name","status"], // subset of proposed_mirrored_attrs
"write_back_attrs": [], // subset of proposed_write_back_attrs
"update_policy": "RealTime",
"message": "Happy to link – sharing name and status only."
}
```

Responses:

Status	Meaning	Response Body
200 OK	Consent recorded	Updated HypergraphEdge with consent_status=Accepted Declined. If Accepted, shadow cell provisioning begins asynchronously.
400	Invalid decision	Decision must be accept or decline.
403	Not the target owner	Only the target node owner can respond to consent.
404	Edge not found	

Note:
When consent is Accepted, the shadow cell provisioning pipeline begins. The shadow_cell_id field on the edge will be populated once provisioning completes (poll or use Web

Status	Meaning	Response Body
Socket stream).		

DELETE	/api/v1/graph/edges/{edge_id}	
--------	-------------------------------	--

Delete a HypergraphEdge. For CrossGridLink edges with an active shadow cell, triggers the shadow cell disconnection protocol: shadow_status set to Disconnected. Historical sync data preserved in EventLog.

Parameters:

Name	In	Type	Required	Description
edge_id	path	uuid	Yes	Edge identifier.

Responses:

Status	Meaning	Response Body
204 No Content	Deleted	No body.
403	Insufficient tier	Requires Owner tier of this edge.

Note: After deletion, existing ShadowRows from this edge remain readable but are marked shadow_status=Disconnected

Status	Meaning	Response Body
nnected. They will not receiv e furthe r updat es.		

12. Shadow Cells

GET	<code>/api/v1/graph/shadow-cells</code>
------------	---

List all ShadowCells in this Grid that were provisioned from linked external Grids via CrossGridLink edges.

Parameters:

Name	In	Type	Required	Description
source_grid_id	query	uuid	No	Filter by source Grid.
shadow_status	query	string	No	Filter: Active Degraded Disconnected.
limit	query	integer	No	Default: 50.
cursor	query	string	No	Pagination cursor.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ data: ShadowCell[], meta: PaginationMeta }

POST	<code>/api/v1/graph/shadow-cells/{shadow_id}/sync</code>
-------------	--

Manually trigger a MirrorAttribute sync for a ShadowCell from its source entity. Forces an immediate fetch of the consented attributes from the source Grid.

Parameters:

Name	In	Type	Required	Description
shadow_id	path	uuid	Yes	ShadowCell identifier.

Responses:

Status	Meaning	Response Body
200 OK	Sync complete	{ shadow_id, synced_attrs: [...], synced_at, shadow_status: 'Active' }
403	Insufficient tier	Requires Editor+ tier.
404	Not found	ShadowCell ID not found.
409	Link revoked	The CrossGridLink has been revoked. Sync not possible.
502	Source unavailable	Source Grid did not respond. shadow_status set to Degraded.

13. Graph Analytics

GET	<code>/api/v1/graph/path</code>
------------	---------------------------------

Find the shortest weighted path between two nodes in the Hypergraph using Dijkstra's algorithm. Edge weight is used as the path cost (lower weight = preferred path).

Parameters:

Name	In	Type	Required	Description
from	query	uuid	Yes	Source node ID.
to	query	uuid	Yes	Target node ID.
edge_type	query	string	No	Filter to specific edge types along path.
max_hops	query	integer	No	Maximum path length. Default: 10.

Responses:

Status	Meaning	Response Body
200 OK	Path found	{ path: [NodeId], edges: [EdgeId], total_weight, hop_count }
404	No path	No path exists between the two nodes with given constraints.

Note: Returns the path as an ordered array of node IDs and

Status	Meaning	Response Body
the edge s traversed. Total weight is the sum of edge weights along the path.		

GET	/api/v1/graph/clusters
-----	------------------------

Detect community clusters in the Hypergraph using the Louvain algorithm. Returns cluster assignments for all nodes within a Space or subgraph.

Parameters:

Name	In	Type	Required	Description
space_id	query	uuid	Yes	Space whose link subgraph to analyze.
edge_type	query	string	No	Only consider specified edge types.
min_cluster_size	query	integer	No	Minimum nodes per cluster. Default: 3.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ clusters: [{ cluster_id, node_ids, modularity_score }], total_clusters, overall_modularity }
422	Graph too large	Space subgraph exceeds 1M nodes. Use sampled approximation or reduce scope.

Note: Cluster detection uses Louvain modu

Status	Meaning	Response Body
200	Success. Centrality scores are computed for each node in the Hypergraph. Results are stored as a computed attribute on each node. For graphs with more than 100K nodes, sampled random-walk approximation is used.	

GET	/api/v1/graph/centrality
-----	--------------------------

Compute centrality scores for nodes in the Hypergraph: betweenness, closeness, and eigenvector (PageRank-style) centrality.

Parameters:

Name	In	Type	Required	Description
space_id	query	uuid	No	Scope to Space subgraph.
metric	query	string	No	betweenness closeness eigenvector all. Default: all.

Name	In	Type	Required	Description
top_n	query	integer	No	Return only top N nodes by centrality. Default: 100. Max: 1000.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ nodes: [{ node_id, betweenness, closeness, eigenvector }], computed_at }
202 Accepted	Computing	For large graphs, computation is async. Returns query_id for polling.

Note: Centrality scores are written back as AI-computed attributes on HyperRow nodes for fast cached access. Full recomputation is triggered on significant graph topology changes.

Part VII — Spaces and Workspaces API

14. Spaces

14.1 Space Object Schema

```
{
  "space_id":      "uuid",
  "space_type":    "Organization",          //
  Personal|Team|Organization|Cooperative|      // Collective|Community|Federation|
  Project|Research|Event|Studio|Factory
  "slug":          "pamoja-capital",
  "name":          "Pamoja Capital Cooperative",
  "namespace_path": "kogi://coop/pamoja-capital/",
  "grid_id":       "uuid",
  "visibility":     "Internal",              //
  Private|Protected|Internal|Public|Unlisted
  "status":        "Active",                // Draft|Active|Paused|Restricted|
  Dissolving|Dissolved|Archived
  "member_count":  47,
  "cube_ids":      ["uuid-1", "uuid-2"],
  "workspace_count": 12,
  "governance": {
    "vote_threshold_pct": 51,
    "quorum_pct":        30,
    "proposal_types":    ["policy", "budget", "member", "constitution"],
    "multisig_required": false
  },
  "treasury_ref":  { "account_id": "kogi-bank-uuid", "balance_summary": "..."},
  "link_node_id":  "uuid",
  "federation_links": ["uuid-fed-1"],
  "created_at":    "2025-06-01T09:00:00Z",
  "updated_at":    "2026-03-20T14:00:00Z"
}
```

GET

/api/v1/spaces

List all Spaces the authenticated caller is a member of, or publicly discoverable Spaces.

Parameters:

Name	In	Type	Required	Description
space_type	query	string	No	Filter by SpaceType. Comma-separated.
visibility	query	string	No	Public Internal All (requires auth). Default: shows accessible Spaces.
name_search	query	string	No	Full-text search on Space name and description.

Name	In	Type	Required	Description
limit	query	integer	No	Default: 50.
cursor	query	string	No	Pagination cursor.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ data: Space[], meta: PaginationMeta }

POST /api/v1/spaces

Create a new Space.

Parameters:

Name	In	Type	Required	Description
X-Idempotency-Key	header	string	No	Idempotency key.

Request Body:

```
{
  "space_type": "Team",
  "slug": "engineering-alpha",           // URL-safe, unique within parent namespace
  "name": "Engineering Alpha Squad",
  "description": "Core platform engineering team",
  "visibility": "Internal",
  "governance": {
    "vote_threshold_pct": 51,
    "quorum_pct": 30
  },
  "parent_space_id": "uuid",             // optional: nest under a parent Space
  "template_id": "uuid",                 // optional: apply a Space template
  "initial_members": [                   // optional: invite initial members
    { "user_id": "uuid", "role": "Manager" }
  ]
}
```

Responses:

Status	Meaning	Response Body
201 Created	Space created	Space object
400	Validation error	Invalid slug, missing required fields.
409	Slug taken	A Space with this slug already exists.

Status	Meaning	Response Body
Note: Creating a Space provisions: a root Hypercube for the Space, a KLNK LinkNode for the Space in the Hypergraph, a Namespace entry at the declared namespace_path, and a system Workspace template.		

GET	/api/v1/spaces/{space_id}
-----	---------------------------

Get full Space details including member roster, cube list, and governance config.

Parameters:

Name	In	Type	Required	Description
space_id	path	uuid	Yes	Space identifier.
include_members	query	boolean	No	Include full member roster. Default: false.
include_cubes	query	boolean	No	Include cube list. Default: false.

Responses:

Status	Meaning	Response Body
200 OK	Success	Space object
403	Not a member	Caller is not a member and Space is not Public.
404	Not found	

PATCH `/api/v1/spaces/{space_id}`

Update Space configuration: name, description, visibility, governance config.

Parameters:

Name	In	Type	Required	Description
space_id	path	uuid	Yes	Space identifier.

Request Body:

```
{
  "name": "Engineering Alpha (Platform)",
  "description": "Updated description",
  "visibility": "Public",
  "governance": {
    "vote_threshold_pct": 66
  }
}
```

Responses:

Status	Meaning	Response Body
200 OK	Updated	Updated Space object
403	Insufficient tier	Requires Manager+ tier within this Space.
409	Governance required	Visibility changes may require a GovernanceProposal vote.

14.2 Space Membership

GET**/api/v1/spaces/{space_id}/members**

List Space members with roles, join dates, and contribution weights.

Parameters:

Name	In	Type	Required	Description
space_id	path	uuid	Yes	Space identifier.
role	query	string	No	Filter by role: Owner Governor Steward Treasurer Editor Contributor Member Viewer.
limit	query	integer	No	Default: 100.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ data: [{ user_id, identity_tag, role, joined_at, contribution_weight, stake_pct }], meta: PaginationMeta }
403	Not a member	Requires Member tier.

POST**/api/v1/spaces/{space_id}/members**

Add a new member to a Space or accept a pending invitation.

Parameters:

Name	In	Type	Required	Description
space_id	path	uuid	Yes	Space identifier.

Request Body:

```
{
  "user_id":      "uuid",
  "identity_tag": "@alice-dev",
  "role":         "Contributor",           // role within the Space
  "message":      "Welcome to the team!"
}
```

Responses:

Status	Meaning	Response Body
201 Created	Member added	SpaceMember object
403	Insufficient tier	Requires Steward+ tier to add members.
409	Already a member	User is already a member of this Space.

PATCH `/api/v1/spaces/{space_id}/members/{user_id}`

Update a member's role within a Space.

Parameters:

Name	In	Type	Required	Description
space_id	path	uuid	Yes	Space identifier.
user_id	path	uuid	Yes	Member user ID.

Request Body:

```
{ "role": "Manager" }
```

Responses:

Status	Meaning	Response Body
200 OK	Updated	Updated SpaceMember object
403	Insufficient tier	Requires Steward+ tier to change member roles.

DELETE `/api/v1/spaces/{space_id}/members/{user_id}`

Remove a member from a Space.

Parameters:

Name	In	Type	Required	Description
space_id	path	uuid	Yes	Space identifier.
user_id	path	uuid	Yes	Member user ID.

Responses:

Status	Meaning	Response Body
204 No Content	Removed	No body.
403	Insufficient tier	Requires Manager+ tier to remove members. Members can remove themselves.

14.3 Space Governance

GET `/api/v1/spaces/{space_id}/governance/proposals`

List governance proposals for a Space.

Parameters:

Name	In	Type	Required	Description
space_id	path	uuid	Yes	Space identifier.
status	query	string	No	open passed rejected all. Default: open.
limit	query	integer	No	Default: 20.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ data: GovernanceProposal[], meta }

POST /api/v1/spaces/{space_id}/governance/proposals

Submit a governance proposal to a Space.

Parameters:

Name	In	Type	Required	Description
space_id	path	uuid	Yes	Space identifier.

Request Body:

```
{
  "title": "Increase vote threshold to 66%",
  "description": "To better represent minority voices, we propose raising the voting threshold.",
  "proposal_type": "policy",
  "action": {
    "type": "update_governance",
    "changes": { "vote_threshold_pct": 66 }
  },
  "voting_deadline": "2026-04-15T23:59:59Z"
}
```

Responses:

Status	Meaning	Response Body
201 Created	Proposal submitted	GovernanceProposal object with status=Open
403	Insufficient tier	Requires Manager+ tier.

POST**/api/v1/spaces/{space_id}/governance/proposals/{proposal_id}/vote**

Cast a vote on an open governance proposal.

Parameters:

Name	In	Type	Required	Description
space_id	path	uuid	Yes	Space identifier.
proposal_id	path	uuid	Yes	Proposal identifier.

Request Body:

```
{
  "vote": "yes", // yes | no | abstain
  "comment": "I support this change."
}
```

Responses:

Status	Meaning	Response Body
200 OK	Vote cast	{ proposal_id, your_vote, current_tally: { yes, no, abstain, quorum_reached, threshold_met } }
400	Voting closed	Voting deadline has passed or proposal is no longer open.
409	Already voted	Caller has already voted on this proposal.

15. Workspaces

POST**/api/v1/workspaces**

Create a new Workspace within a Space.

Parameters:

Name	In	Type	Required	Description
X-Idempotency-Key	header	string	No	Idempotency key.

Request Body:

```
{
  "name": "Q2 Planning Workspace",
  "space_id": "uuid", // required
  "workspace_type": "Personal", // Personal|Shared|Project|Template
  "description": "Planning session for Q2 2026",
  "initial_config": {
    "open_cubes": [
      { "cube_id": "uuid", "view_id": "uuid" }
    ]
  }
}
```

```

    },
    "dim_context": {
      "time_axis_id": "2026-Q2"           // default time slice for all cubes
    }
  }
}

```

Responses:

Status	Meaning	Response Body
201 Created	Workspace created	Workspace object
403	Insufficient tier	Requires Editor+ tier.

GET `/api/v1/workspaces/{workspace_id}`

Get current Workspace state including open cubes, dim context, and active session info.

Parameters:

Name	In	Type	Required	Description
workspace_id	path	uuid	Yes	Workspace identifier.

Responses:

Status	Meaning	Response Body
200 OK	Success	Workspace object with full session state
403	Not a member	Caller is not a member of the Space owning this Workspace.

PUT `/api/v1/workspaces/{workspace_id}/dim-context`

Set the workspace-level DimensionKey defaults. When set, ALL open cubes in this Workspace automatically show the specified dimension slice. This is the N-dimensional generalization of freeze panes.

Parameters:

Name	In	Type	Required	Description
workspace_id	path	uuid	Yes	Workspace identifier.

Request Body:

```
{
```

```
"dim_context": {
  "time_axis_uuid":      "2026-Q2",      // all cubes default to Q2 time slice
  "geo_axis_uuid":       "EMEA",         // all cubes default to EMEA geo slice
  "scenario_axis_uuid":  "base_case"     // all cubes show base case scenario
}
```

Responses:

Status	Meaning	Response Body
200 OK	Updated	Updated Workspace object
400	Invalid axis ID	One or more axis IDs not found in any open cube.

Note: Setting a dim_context does not modify any cell data. It adds a DimSlice predicate to all queries executed within this Workspace session, providing a coherent multi-cube view of the same

Status	Meaning	Response Body
dime nsion al slice.		

POST	/api/v1/workspaces/{workspace_id}/session
------	---

Open a Workspace session. Returns a WebSocket URL for real-time collaborative editing and a session token.

Parameters:

Name	In	Type	Required	Description
workspace_id	path	uuid	Yes	Workspace identifier.

Request Body:

```
{ "client_info": { "platform": "web", "user_agent": "..." } }
```

Responses:

Status	Meaning	Response Body
200 OK	Session opened	{ session_id, websocket_url, session_token, expires_at }
403	Not a member	

Note:
Use
the
webs
ocket
_url
to
conn
ect
to
the
collab
orativ
e
editin
g
Web
Sock
et
chan
nel.
The
sessi

Status	Meaning	Response Body
on_token is required as a query parameter : ?session_token={token}. Sessions expire after 1 hour of inactivity.		

Part VIII — Namespace API

16. Namespace Registry

The Namespace system provides globally unique, hierarchical URI addressing for every entity. The NamespacePath scheme is:
hypergrid://{grid_name}/{space_type}/{slug}/{entity_type}/{entity_slug}/... for Hypergrid-native entities, and domain-specific prefixes (kogi://, ume://, qala://) for domain system entities.

GET	/api/v1/ns/resolve
-----	--------------------

Resolve a NamespacePath URI to its entity ID, entity type, and Grid location. Supports aliases and federated entries.

Parameters:

Name	In	Type	Required	Description
path	query	string	Yes	NamespacePath URI to resolve. URL-encoded. Example: kogi://coop/pamoja-capital/
include_metadata	query	boolean	No	Include full NamespaceEntry metadata. Default: false.

Responses:

Status	Meaning	Response Body
200 OK	Resolved	{ entity_id: 'uuid', entity_type: 'Space', grid_id: 'uuid', namespace_path: '...', canonical_url: '...', resolved_from: 'direct alias federated' }
404	Not found	Path does not resolve to any known entity. This path may not have been registered.
302 / redirect_url	Alias	When the path is an alias, returns redirect_url pointing to the canonical path.

Note: Namespace resolution is Redis-cached with a 300s TTL. The

Status	Meaning	Response Body
response includes resolved_from to indicate whether this was a direct match, alias redirect, or federated lookup from a peer Grid.		

POST	/api/v1/ns
------	------------

Register a new NamespacePath entry, binding it to an entity.

Parameters:

Name	In	Type	Required	Description
X-Idempotency-Key	header	string	No	Idempotency key.

Request Body:

```
{
  "path": "kogi://coop/pamoja-capital/projects/webapp-2026/",
  "entity_id": "uuid",
  "entity_type": "Row",
  "visibility": "Public",
  "canonical_url": "https://kogi.io/coop/pamoja-capital/projects/webapp-2026"
}
```

Responses:

Status	Meaning	Response Body
201 Created	Registered	NamespaceEntry object
400	Invalid path	Path format is invalid or conflicts with system-reserved paths.
403	Insufficient tier	Must be Owner of the parent namespace segment.
409	Already exists	A namespace entry at this path already exists.
Note: Name space regist ration is propa gated to all feder ation peers auto matic ally. Cross -linke d entitie s beco me resolv able at all peer Grids once regist ered.		

GET	/api/v1/ns/list
-----	-----------------

List namespace entries under a path prefix.

Parameters:

Name	In	Type	Required	Description
prefix	query	string	Yes	Namespace path prefix. Example: kogi://coop/pamoja-capital/

Name	In	Type	Required	Description
depth	query	integer	No	Maximum depth below prefix. Default: 1 (direct children only). Max: 5.
entity_type	query	string	No	Filter by entity type.
limit	query	integer	No	Default: 100. Max: 1000.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ data: NamespaceEntry[], meta: PaginationMeta }

POST `/api/v1/ns/{namespace_id}/alias`

Create an additional path alias that resolves to the same entity as the canonical path.

Parameters:

Name	In	Type	Required	Description
namespace_id	path	uuid	Yes	NamespaceEntry identifier.

Request Body:

```
{ "alias_path": "kogi://webapp-2026/" }
```

Responses:

Status	Meaning	Response Body
201 Created	Alias created	{ alias_path, canonical_path, namespace_id }
403	Insufficient tier	Requires Owner of this namespace entry.
409	Already exists	Alias path is already in use.

PATCH `/api/v1/ns/{namespace_id}`

Rename a namespace path segment. Creates an alias from the old path to the new path, so existing references continue to work.

Parameters:

Name	In	Type	Required	Description
namespace_id	path	uuid	Yes	NamespaceEntry identifier.

Request Body:

```
{
  "path": "kogi://coop/pamoja-capital/projects/webapp-v2-2026/",
  "create_alias": true // create alias from old path (default: true)
}
```

Responses:

Status	Meaning	Response Body
200 OK	Renamed	Updated NamespaceEntry
403	Insufficient tier	Requires Owner of this namespace entry.
409	Already exists	New path is already in use.

POST `/api/v1/ns/{namespace_id}/federate`

Replicate a namespace entry to a federation peer Grid, making it resolvable at the peer.

Parameters:

Name	In	Type	Required	Description
namespace_id	path	uuid	Yes	NamespaceEntry identifier.

Request Body:

```
{ "peer_grid_id": "uuid" }
```

Responses:

Status	Meaning	Response Body
200 OK	Federated	{ namespace_id, federated_to: GridId[] }
403	Insufficient tier	Requires Admin tier.
404	Peer not found	Peer Grid not registered as a federation peer.

GET `/api/v1/ns/health`

Get namespace health report: stale entries, broken aliases, unresolvable paths, federation sync status.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ total_entries, stale_count, broken_aliases, avg_resolve_latency_ms, federation_sync_status: { peer_id, last_sync_at, pending_count }[] }
403	Admin required	Requires Admin tier.

Part IX — Identity and Access API

17. Identity Management

GET `/api/v1/identity/me`

Get the authenticated caller's identity profile: *SovereignEntity*, all *TenantPartitions*, active spaces and workspaces.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ sovereign_entity_id, partitions: TenantPartition[], active_spaces: Space[], api_keys: [{ key_id, name, scopes, last_used }] }

GET `/api/v1/identity/partitions`

List all *TenantPartitions* (identity handles) for the authenticated *SovereignEntity*.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ data: TenantPartition[] }

POST `/api/v1/identity/partitions`

Create a new *TenantPartition* (a new named identity handle) for the authenticated *SovereignEntity*.

Parameters:

Name	In	Type	Required	Description
X-Idempotency-Key	header	string	No	Idempotency key.

Request Body:

```
{
  "identity_handle": "@alice-creative",      // @handle - unique within
  SovereignEntity
  "display_name": "Alice (Creative Work)",
  "isolation_level": "SoftIsolated",        // None|SoftIsolated|HardIsolated
  "visibility_mask": {
    "public_attrs": ["name", "bio", "tags", "portfolio_url"],
    "follower_attrs": ["name", "bio", "tags", "portfolio_url", "recent_projects"],
    "owner_only_attrs": ["bank_account_ref", "tax_id", "ssn_ref"]
  }
}
```

```
}  
}
```

Responses:

Status	Meaning	Response Body
201 Created	Partition created	TenantPartition object
400	Invalid handle	Handle contains invalid characters or is reserved.
409	Handle taken	An identity handle with this name already exists.

PATCH `/api/v1/identity/partitions/{partition_id}`*Update a TenantPartition: display name, isolation level, visibility mask.***Parameters:**

Name	In	Type	Required	Description
partition_id	path	uuid	Yes	Partition identifier.

Request Body:

```
{  
  "display_name":    "Alice - Developer Identity",  
  "isolation_level": "HardIsolated",  
  "visibility_mask": {  
    "public_attrs": ["name", "bio"]  
  }  
}
```

Responses:

Status	Meaning	Response Body
200 OK	Updated	Updated TenantPartition
400	Cannot downgrade isolation	HardIsolated cannot be downgraded without explicit force flag.

POST `/api/v1/identity/partitions/merge`*Initiate a CrossIdentityMerge: merge two TenantPartitions of the same SovereignEntity. Irreversible for KLNK node merging. Full checkpoint created before merge begins.***Request Body:**

```
{
```

```
"source_partition_id": "uuid",           // partition to merge FROM (will be archived)
"target_partition_id": "uuid",           // partition to merge INTO (survives)
"dry_run": true,                         // simulate without executing
(default: true)
"create_namespace_alias": true           // create redirect from source @handle to target
}
```

Responses:

Status	Meaning	Response Body
200 OK	Dry run complete	{ source_handle, target_handle, rows_to_retag, shared_rows, namespace_aliases_to_create, klnk_edges_to_merge, checkpoint_size_estimate_mb }
202 Accepted	Merge initiated (dry_run=false)	{ merge_id, checkpoint_id, status: 'in_progress', estimated_completion_at }
400	Cannot merge	Source and target belong to different SovereignEntities.

Note: Always perform a dry run first to review the impact. The actual merge creates a pre-merge snapshot (restorable within 7 days)

· KLNK node merging is perm

Status	Meaning	Response Body
201 Created	API key created	{ key_id, name, api_key: '...', scopes, expires_at }. NOTE: api_key value shown ONCE — store securely.

POST	/api/v1/identity/api-keys
------	---------------------------

Create a new API Key for server-to-server access.

Parameters:

Name	In	Type	Required	Description
X-Idempotency-Key	header	string	No	Idempotency key.

Request Body:

```
{
  "name": "kogi-engine-service-key",
  "scopes": ["cubes:read","cells:write:ai","eventlog:append"],
  "cube_restrictions": ["uuid-cube-1", "uuid-cube-2"], // empty = all cubes
  "expires_at": "2027-01-01T00:00:00Z", // null = non-expiring
  "description": "Used by kogi-engine WritebackService"
}
```

Responses:

Status	Meaning	Response Body
201 Created	API key created	{ key_id, name, api_key: '...', scopes, expires_at }. NOTE: api_key value shown ONCE — store securely.
400	Invalid scope	One or more scopes are not recognized.

Note: The api_key secret value is shown only in the 201 response. It cannot be retrieved

Status	Meaning	Response Body
again . If lost, delet e and recre ate the key.		

DELETE	/api/v1/identity/api-keys/{key_id}
--------	------------------------------------

Revoke an API Key immediately.

Parameters:

Name	In	Type	Required	Description
key_id	path	uuid	Yes	API key identifier.

Responses:

Status	Meaning	Response Body
204 No Content	Revoked	No body. All in-flight requests using this key will fail immediately.

Part X — AI and Intelligence API

18. AI Computation

POST	/api/v1/ai/compute
------	--------------------

Request Tier-2 AI-computed attribute values for a batch of entities. Triggers `AIEngineAdapter.compute_batch()` on the specified engine. Results are written back to `HyperCells` via `WritebackService` and returned synchronously if available within timeout.

Request Body:

```
{
  "engine_id": "kogi.health_score_engine.v2",
  "cube_id": "uuid",
  "entity_ids": ["uuid-1","uuid-2","uuid-3"],
  "force": false, // force recompute even if within cache TTL
  "sync": true, // wait for results (up to timeout_ms)
  "timeout_ms": 5000 // max wait. if exceeded, returns job_id for polling
}
```

Responses:

Status	Meaning	Response Body
200 OK	Results ready (sync=true, within timeout)	{ results: [{ entity_id, outputs: { attr_key: value, ... }, confidence, computed_at }] }
202 Accepted	Computing (sync=false or timed out)	{ job_id, status: 'queued', poll_url: '...' }
400	Engine not found	engine_id not registered.
403	Insufficient tier	Requires Editor+ tier.

Note: AI computation is billed per-entity per-compute. Use force=false

Status	Meaning	Response Body
(default) to serve cached results within the engine's configured staleness_ttl.		

GET `/api/v1/ai/anomalies`

List AI-detected anomalies in a cube or DimSlice. Anomalies are detected by registered ALEngineAdapter plugins via `detect_anomalies()`.

Parameters:

Name	In	Type	Required	Description
cube_id	query	uuid	Yes	Hypercube to scan.
d1_key	query	uuid	No	Scope to a specific HyperRow.
attr_key	query	string	No	Scope to a specific attribute key.
severity_min	query	number	No	Minimum anomaly severity (0.0–1.0). Default: 0.3.
kind	query	string	No	SuddenChange PatternBreak OutlierValue StateChange
since	query	datetime	No	Only anomalies detected after this timestamp.
limit	query	integer	No	Default: 50.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ data: [{ anomaly_id, entity_id, attr_key, kind, severity, baseline, deviation, detected_at }], meta: PaginationMeta }

POST `/api/v1/ai/anomalies/{anomaly_id}/resolve`

Mark an anomaly as acknowledged or resolved. Does not delete the anomaly record from the EventLog.

Parameters:

Name	In	Type	Required	Description
anomaly_id	path	uuid	Yes	Anomaly identifier.

Request Body:

```
{
  "resolution": "acknowledged",           // acknowledged | resolved |
  false_positive
  "note": "Intentional budget increase — Q2 push."
}
```

Responses:

Status	Meaning	Response Body
200 OK	Resolved	Updated anomaly record

GET `/api/v1/ai/engines`

List all registered *AIEngineAdapter* plugins and their health status.

Parameters:

Name	In	Type	Required	Description
status	query	string	No	Healthy Degraded Unhealthy — filter by health status.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ data: [{ engine_id, name, version, capabilities, health, output_attrs, cache_hit_rate }] }
403	Admin required	

19. WritebackService — AI Engine Write Protocol

The WritebackService is the gRPC endpoint used exclusively by AI engines to write computed attribute values into HyperCells. It is not accessible over REST from external clients. The following documents the protocol for implementors building *AIEngineAdapter* plugins that write back to Hypergrid.

```
// WritebackService gRPC interface (proto3)
service WritebackService {
  rpc WriteColumn(WriteColumnRequest) returns (WriteColumnResponse);
  rpc WriteBatch(WriteBatchRequest) returns (WriteBatchResponse);
  rpc WriteAnomalyFlag(WriteAnomalyRequest) returns (WriteAnomalyResponse);
}
```

```

}

message WriteColumnRequest {
    string grid_id      = 1;
    string cube_id      = 2;
    string dl_key       = 3;    // entity D: key
    string attr_key     = 4;    // must be attr_type=AI or System
    google.protobuf.Value value = 5;
    float confidence    = 6;    // 0.0-1.0
    string source_engine = 7;    // registered engine_id
    int64 computation_ts = 8;    // nanosecond epoch
    int64 ttl_seconds    = 9;    // Redis CellCache TTL
    string model_version = 10;
    string identity_tag  = 11;   // optional partition scope
    string space_id      = 12;   // optional Space scope
    // Optional: N-dim coordinates for higher dimensions
    repeated DimKeyValue extra_dims = 13;
}

message WriteBatchRequest {
    repeated WriteColumnRequest writes = 1;
    bool atomic = 2;                  // if true: all or nothing
}

// Processing pipeline (7 steps):
// 1. Validate source_engine registered and has AIWriter permission for attr_key
// 2. Evaluate PolicyEngine for blocking policies
// 3. Threshold check: anomaly alert if value crosses configured threshold
// 4. Write as SetAttr CrdtOp with actor=ai::{source_engine}::{model_version}
// 5. Update Redis CellCache with TTL
// 6. Append EventEntry::AIColumnUpdated to EventLog
// 7. WebSocket broadcast CellUpdateEvent to subscribed Workspaces

```

20. Oba AI Assistant API

Oba is the domain-specific AI assistant layer built on top of the Hypergrid AI infrastructure. It provides reactive query answering, proactive anomaly surfacing, and approval-first autonomous action preparation.

POST **/api/v1/ai/oba/query**

Submit a natural language query to Oba. Returns both AI-generated insights and structured data from HyperQL execution.

Request Body:

```

{
  "question":    "Show me everything at risk this week in my engineering
portfolio",
  "cube_context":["kogi.portfolio.components"],
  "identity_tag":"@alice-dev",
  "space_id":    "uuid",
  "return_hyperql": true,
  "return_insights": true           // AI narrative in addition to data
}

```

```
}

```

Responses:

Status	Meaning	Response Body
200 OK	Success	{ answer: string, generated_hyperql: string, columns, rows, insights: [{ type, message, affected_rows }], suggestions: [ObaSuggestion] }

GET `/api/v1/ai/oba/hints/{entity_id}`

Get pending Oba proactive hints for a specific entity: anomaly warnings, completion suggestions, upcoming deadlines, risk flags.

Parameters:

Name	In	Type	Required	Description
entity_id	path	uuid	Yes	Entity (HyperRow D1 key).
cube_id	query	uuid	Yes	Cube containing the entity.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ entity_id, hints: [{ hint_id, type, message, severity, suggested_action, expires_at }] }

POST `/api/v1/ai/oba/actions`

Submit an Oba autonomous action for user approval. The action is described in structured form with a human-readable preview. No mutations occur until the user confirms via `POST /ai/oba/actions/{action_id}/confirm`.

Request Body:

```
{
  "action_type": "batch_archive",
  "description": "Archive 7 completed projects older than 90 days",
  "preview": {
    "affected_entities": ["uuid-1","uuid-2","..."],
    "hyperql_preview": "UPDATE components SET status='Archived' WHERE ...",
    "reversible": true
  },
  "cube_id": "uuid",
  "expires_at": "2026-03-23T17:00:00Z"
}
```

Responses:

Status	Meaning	Response Body
201 Created	Action staged	{ action_id, status: 'awaiting_confirmation', preview, expires_at }
Note: Obas have a configurable expiry window. If not confirmed before expires_at, the action is discarded without any mutations.		

POST	/api/v1/ai/oba/actions/{action_id}/confirm
------	--

Confirm and execute a pending Oba autonomous action. This is the only way to execute an Oba action — no autonomous execution occurs without explicit user confirmation.

Parameters:

Name	In	Type	Required	Description
action_id	path	uuid	Yes	Oba action identifier.

Request Body:

```
{ "confirmed": true }
```

Responses:

Status	Meaning	Response Body
200 OK	Action executed	{ action_id, status: 'completed', mutations_applied, entities_affected }
400	Action expired	Action has expired. Submit a new action request.
404	Not found	Action ID not found.
Note: All mutations from a confirmed Oba action are recorded in the Event Log with actor =oba: {action_id}. They can be reversed via POST /ai/oba/actions/{action_id}/undo within the undo window (default: 30 minutes).		

Part XI — Federation API

21. Federation Peers

GET `/api/v1/federation/peers`

List all registered federation peers for this Grid.

Parameters:

Name	In	Type	Required	Description
status	query	string	No	Active Degraded Disconnected — filter by peer status.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ data: [{ peer_id, grid_id, peer_url, trust_level, status, lag_ms, last_heartbeat_at, schema_aligned }] }
403	Admin required	

POST `/api/v1/federation/peers`

Register a new federation peer. Initiates the full 8-phase federation handshake.

Parameters:

Name	In	Type	Required	Description
X-Idempotency-Key	header	string	No	Idempotency key.

Request Body:

```
{
  "peer_url": "https://ume.hypergrid.io",
  "peer_grid_id": "uuid",
  "trust_level": "grid", // grid|namespace|identity
  "description": "Ume B-OS Grid"
}
```

Responses:

Status	Meaning	Response Body
201 Created	Peer registered	{ peer_id, grid_id, peer_url, trust_level, status: 'handshaking' }

Status	Meaning	Response Body
400	Invalid peer URL	Peer URL is unreachable or does not respond to federation handshake.
403	Admin required	
409	Already registered	This peer Grid is already registered.
Note: Federation registration initiates the 8-phase handshake asynchronously. Poll GET /federation/peers to track progress. Phases: Discovery → Handshake → Schema Sync → Cell Delta Sync → Conflict Surface → Namespace Sync →		

Status	Meaning	Response Body
Heart beat → Shadow Sync.		

GET `/api/v1/federation/peers/{peer_id}`

Get detailed status for a specific federation peer: VectorClock lag, schema alignment, active shadow syncs.

Parameters:

Name	In	Type	Required	Description
peer_id	path	uuid	Yes	Federation peer identifier.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ peer_id, status, vector_clock_lag, schema_version_local, schema_version_remote, unsynced_ops_count, active_shadow_syncs, last_sync_at }

POST `/api/v1/federation/sync`

Trigger a manual full delta sync with all registered federation peers. Used to force synchronization after a network partition or maintenance window.

Request Body:

```
{ "peer_id": "uuid" }
```

Responses:

Status	Meaning	Response Body
202 Accepted	Sync initiated	{ sync_id, status: 'running', peers_syncing: [uuid] }
403	Admin required	

GET `/api/v1/federation/conflicts`

List unresolved CRDT conflicts that require human review. Conflicts arise when concurrent writes from different federation nodes cannot be automatically merged.

Parameters:

Name	In	Type	Required	Description
cube_id	query	uuid	No	Filter to conflicts in a specific Hypercube.
entity_id	query	uuid	No	Filter to conflicts for a specific entity.
resolved	query	boolean	No	Include resolved conflicts. Default: false.
limit	query	integer	No	Default: 50.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ data: [{ conflict_id, cube_id, entity_id, attr_key, value_a, value_b, node_a, node_b, detected_at, resolved }] }
403	Editor required	

POST `/api/v1/federation/conflicts/{conflict_id}/resolve`

Resolve a CRDT conflict by choosing a resolution strategy: *keep_local*, *keep_remote*, *merge* (JSON CRDT), or provide a custom value.

Parameters:

Name	In	Type	Required	Description
conflict_id	path	uuid	Yes	Conflict identifier.

Request Body:

```
{
  "resolution": "keep_remote",           // keep_local|keep_remote|merge|custom
  "custom_value": null,                 // required if resolution=custom
  "note": "Remote value is correct - was an accidental local edit."
}
```

Responses:

Status	Meaning	Response Body
200 OK	Resolved	{ conflict_id, resolved_value, resolved_by, resolved_at }
404	Not found	Conflict ID not found.

Part XII — Export, Snapshots, and EventLog

22. Export

GET	/api/v1/cubes/{cube_id}/export
-----	--------------------------------

Export a Hypercube or HypercubeView in the requested format. Supports CSV, XLSX, JSON, JSON-LD, Apache Parquet, and Apache Arrow. For large cubes, an async job is returned.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.
format	query	string	Yes	csv xlsx json jsonld parquet arrow
view_id	query	uuid	No	Apply a saved HypercubeView (filters, sorts, dim slices).
filter	query	string	No	Inline DimSlice filter in HyperQL WHERE syntax.
attrs	query	string	No	Comma-separated attribute keys to export. Default: all.
as_of	query	datetime	No	Export historical snapshot at this timestamp.
include_schema	query	boolean	No	Include cube schema in export. Default: true for json/jsonld/parquet.

Responses:

Status	Meaning	Response Body
200 OK	Streaming download (< 50MB)	Binary file stream with appropriate Content-Type and Content-Disposition headers.
202 Accepted	Async export queued (>= 50MB)	{ export_id, status: 'queued', estimated_size_mb, download_url_expires_in: 3600 }

Note: For N>2 cubes, CSV and XLSX exports automatically apply DimF

Status	Meaning	Response Body
old to reduce to 2D. Set view_id to control which DimFolds are applied. JSON and Parquet preserve the full N-dimensional structure.		

GET	/api/v1/export/{export_id}
-----	----------------------------

Poll the status of an async export job and get the download URL when complete.

Parameters:

Name	In	Type	Required	Description
export_id	path	uuid	Yes	Export job identifier.

Responses:

Status	Meaning	Response Body
200 OK	Pending	{ export_id, status: 'running', progress_pct }
200 OK	Complete	{ export_id, status: 'complete', download_url, size_bytes, expires_at }
200 OK	Failed	{ export_id, status: 'failed', error }

Status	Meaning	Response Body
Note: Down load URLs are pre-si gned S3 URLs valid for 1 hour. The expor t file is retain ed for 24 hours after gener ation.		

23. Snapshots

POST	/api/v1/cubes/{cube_id}/snapshot
------	----------------------------------

Create a named point-in-time snapshot of a Hypercube. Snapshots enable efficient AS_OF time-travel queries and disaster recovery.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.

Request Body:

```
{
  "label":      "before-q2-migration",    // human-readable label for this
  snapshot
  "description": "Pre-migration snapshot before Q2 2026 data import",
  "retain_until": "2026-12-31T23:59:59Z" // null = use default retention (90
  days)
}
```

Responses:

Status	Meaning	Response Body
201 Created	Snapshot queued	{ snapshot_id, cube_id, label, status: 'queued', estimated_size_mb }
403	Owner required	
Note: Snapshot creation is asynchronous. Large cubes may take several minutes. Poll GET /cubes/{id}/snapshots/{snapshot_id} for status.		

GET	/api/v1/cubes/{cube_id}/snapshots
-----	-----------------------------------

List all snapshots for a Hypercube.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.
status	query	string	No	complete pending failed — filter by status.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ data: [{ snapshot_id, label, status, size_bytes, created_at, retain_until }] }

POST	/api/v1/cubes/{cube_id}/restore/{snapshot_id}
------	---

Restore a Hypercube from a snapshot. All cell data is rolled back to the snapshot state. EventLog is preserved — the restore itself is recorded as a RestoreEvent.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.
snapshot_id	path	uuid	Yes	Snapshot to restore from.

Request Body:

```
{
  "dry_run": true,                                // simulate without executing (default: true)
  "confirm": "RESTORE"                            // required when dry_run=false: type 'RESTORE'
}
```

Responses:

Status	Meaning	Response Body
200 OK	Dry run result	{ rows_affected, attrs_rolled_back, estimated_duration_ms }
202 Accepted	Restore initiated	{ restore_id, status: 'running', estimated_completion_at }
400	Confirmation required	dry_run=false requires confirm='RESTORE'.
403	Owner required	
404	Snapshot not found	Snapshot expired or ID not found.

Note: Restoration creates a checkpoint of the current state before rolling back (restorable)

Status	Meaning	Response Body
within 7 days) . The Event Log records the full restore as a single RestoreEvent, preserving the complete audit trail of all mutations since the snapshot.		

24. EventLog

GET	/api/v1/cubes/{cube_id}/eventlog
-----	----------------------------------

Query the EventLog for a Hypercube. All mutations, schema changes, AI writebacks, governance actions, and CRDT conflicts are recorded here. Immutable — no entry can be modified or deleted.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.
entity_id	query	uuid	No	Filter to events for a specific HyperRow (D _i key).
attr_key	query	string	No	Filter to events affecting a specific attribute key.
event_type	query	string	No	Filter by EventEntry type. Comma-separated.

Name	In	Type	Required	Description
actor	query	string	No	Filter by actor ID (user, AI engine, federation node).
since	query	datetime	No	Events after this timestamp.
until	query	datetime	No	Events before this timestamp.
limit	query	integer	No	Default: 100. Max: 1000.
cursor	query	string	No	Pagination cursor. EventLog is paginated newest-first.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ data: EventLogEntry[], meta: PaginationMeta }
403	Editor required	

24.1 EventLogEntry Object Schema

```
{
  "entry_id":      "uuid",
  "cube_id":      "uuid",
  "entity_id":    "uuid",           // D: key of affected entity
  "attr_key":     "status",        // affected attribute key
  "event_type":   "SetAttr",       //
  SetAttr|AddToSet|RemoveFromSet|IncrCounter|
                                     //
  TransitionState|AppendLog|SchemaChange|
                                     //
  AIColumnUpdated|AnomalyDetected|Restored|
                                     //
  ShadowSync|FederationSync|GovernanceAction
  "before_value": { "value": "Draft", "type": "Enum" },
  "after_value":  { "value": "Active", "type": "Enum" },
  "delta":        { "op": "SetAttr", "attr_key": "status", "value": "Active", ...
},
  "vector_clock": { "node-us-east-1": 42 },
  "actor":        "user-uuid:@alice-dev:node-us-east-1",
  "identity_tag": "@alice-dev",
  "space_id":     "uuid",
  "workspace_id": "uuid",
  "created_at":   "2026-03-22T10:00:00Z",
  "is_conflict":  false
}
```

Part XIII — WebSocket and Streaming API

25. WebSocket Channels

Hypergrid's WebSocket API provides real-time cell mutation streaming, collaborative editing, Space activity feeds, shadow cell synchronization, and CRDT federation gossip. All WebSocket connections use WSS (WebSocket over TLS). Authentication is via `session_token` query parameter (returned by `POST /workspaces/{id}/session`).

25.1 Cube Stream — Real-Time Cell Mutations

Attribute	Value
URL	<code>wss://{grid_host}/ws/v1/cubes/{cube_id}/stream</code>
Auth	<code>?session_token={token}</code> or <code>?api_key={key}</code>
Protocol	JSON messages over WSS
Message direction	Server → Client (subscribe) and Client → Server (subscribe/unsubscribe)

```
// Subscribe to all cells in a cube:
// Client → Server
{ "op": "subscribe", "scope": "cube", "cube_id": "uuid" }

// Subscribe to a specific HyperRow:
{ "op": "subscribe", "scope": "row", "cube_id": "uuid", "d1_key": "uuid" }

// Subscribe to a specific attribute key across all rows:
{ "op": "subscribe", "scope": "attr", "cube_id": "uuid", "attr_key":
"health_score" }

// Subscribe to a DimSlice:
{ "op": "subscribe", "scope": "slice", "cube_id": "uuid",
  "filter": "D2='status' AND value='Active'" }

// Unsubscribe:
{ "op": "unsubscribe", "subscription_id": "sub-uuid" }

// Heartbeat (client must send every 30s to keep connection alive):
{ "op": "ping" }
// Server response:
{ "op": "pong", "server_ts": "2026-03-22T10:00:00Z" }

// — Incoming events (Server → Client) —————
// Cell updated:
{
  "event": "cell_updated",
  "subscription_id": "sub-uuid",
  "cube_id": "uuid",
  "d1_key": "uuid",
```

```
"attr_key": "health_score",
"new_value": { "value": 87.4, "type": "AI", "confidence": 0.92 },
"vector_clock": { "node-us-east-1": 43 },
"actor": "ai::kogi.health_score_engine.v2::v2.1",
"ts": "2026-03-22T10:00:01Z"
}

// Row created:
{ "event": "row_created", "cube_id": "uuid", "dl_key": "uuid", "ts": "..." }

// Row archived:
{ "event": "row_archived", "cube_id": "uuid", "dl_key": "uuid", "ts": "..." }

// Schema event (attr key added):
{ "event": "attr_registered", "cube_id": "uuid", "attr_key": "new_field", "ts": "..." }

// AI writeback:
{ "event": "ai_writeback", "cube_id": "uuid", "dl_key": "uuid",
  "engine": "kogi.health_score_engine.v2", "attr_key": "health_score",
  "value": 91.2, "confidence": 0.95, "ts": "..." }

// CRDT conflict:
{ "event": "conflict_detected", "cube_id": "uuid", "dl_key": "uuid",
  "attr_key": "status", "conflict_id": "uuid", "ts": "..." }
```

25.2 Workspace Session — Collaborative Editing

Attribute	Value
URL	wss://{grid_host}/ws/v1/workspaces/{workspace_id}/session
Auth	?session_token={token} (from POST /workspaces/{id}/session)
Purpose	Real-time collaborative editing: cursor presence, cell locking, CRDT synchronization, Oba AI annotations

```
// — Client → Server —————
// Move cursor:
{ "op": "cursor_move", "cube_id": "uuid", "coord": "dl_key,attr_key", "ts": "..."
}

// Lock a cell for editing:
{ "op": "cell_lock", "cube_id": "uuid", "coord": "dl_key,attr_key" }

// Unlock a cell:
{ "op": "cell_unlock", "cube_id": "uuid", "coord": "dl_key,attr_key" }

// Write a cell (sends CrdtOperation to server):
{ "op": "cell_write", "cube_id": "uuid", "coord": "...", "attr_key": "name",
  "value": "New Name", "crdt_op": "SetAttr", "client_ts": "..." }

// Set workspace DimContext:
{ "op": "set_dim_context", "dim_context": { "time_axis_id": "2026-Q2" } }

// — Server → Client —————
```

```
// Presence: another user joined:
{ "event": "presence_joined", "user_id": "uuid", "identity_tag": "@bob-dev",
  "cursor": { "cube_id": "uuid", "coord": "...", "ts": "..."} }

// Presence: user left:
{ "event": "presence_left", "user_id": "uuid", "ts": "..."}

// Cell locked by another user:
{ "event": "cell_locked", "cube_id": "uuid", "coord": "...",
  "locked_by": "uuid", "locked_by_name": "Bob", "ts": "..."}

// Cell unlocked:
{ "event": "cell_unlocked", "cube_id": "uuid", "coord": "...", "ts": "..."}

// DimContext changed by another user (Shared Workspace):
{ "event": "dim_context_changed", "dim_context": { "time_axis_id": "2026-Q3"},
  "changed_by": "uuid", "ts": "..."}

// Oba AI hint for a row:
{ "event": "oba_hint", "entity_id": "uuid", "hint": {
  "hint_id": "uuid",
  "type": "anomaly_warning",
  "message": "Budget utilization at 94% - 3 days until quarter end.",
  "severity": 0.8,
  "expires_at": "...",
}, "ts": "..."}

// Write conflict notification:
{ "event": "conflict_detected", "conflict_id": "uuid", "cube_id": "uuid",
  "coord": "...", "your_value": "A", "winning_value": "B", "ts": "..."}

// Sync complete acknowledgment (your write was committed):
{ "event": "sync_complete", "client_ts": "...", "server_ts": "...", "version": 55
}
```

25.3 Space Activity Stream

Attribute	Value
URL	wss://{grid_host}/ws/v1/spaces/{space_id}/activity
Auth	?session_token={token}
Purpose	Space-level activity feed: member events, governance actions, treasury events, graph edge changes, shadow cell updates

```
// Events emitted on this channel:
{ "event": "member_joined", "user_id": "uuid", "role": "Contributor", "ts": "..."}
{ "event": "member_left", "user_id": "uuid", "ts": "..."}
{ "event": "member_role_changed", "user_id": "uuid", "old_role": "Contributor",
  "new_role": "Manager", "ts": "..."}
{ "event": "proposal_created", "proposal_id": "uuid", "title": "...", "ts": "..."}
{ "event": "vote_submitted", "proposal_id": "uuid", "current_tally": {...},
  "ts": "..."}
}
```

```
{ "event": "proposal_resolved", "proposal_id": "uuid", "outcome": "passed", "ts": "..."}
{ "event": "edge_created", "edge_id": "uuid", "edge_type": "CrossGridLink", "ts": "..."}
{ "event": "edge_consent_accepted", "edge_id": "uuid", "ts": "..."}
{ "event": "shadow_cell_updated", "shadow_id": "uuid", "attr_key": "...", "ts": "..."}
{ "event": "treasury_event", "event_type": "distribution", "amount": {...}, "ts": "..."}
}
```

25.4 Shadow Sync Stream

Attribute	Value
URL	wss://{grid_host}/ws/v1/graph/shadow-sync
Auth	?session_token={token}
Purpose	Real-time MirrorAttribute delta stream for all ShadowCells in this Grid

```
// Events emitted:
{ "event": "shadow_attr_updated", "shadow_id": "uuid", "source_grid": "uuid", "attr_key": "status", "new_value": {...}, "ts": "..."}
{ "event": "shadow_connected", "shadow_id": "uuid", "edge_id": "uuid", "ts": "..."}
{ "event": "shadow_disconnected", "shadow_id": "uuid", "reason": "link_revoked", "ts": "..."}
{ "event": "shadow_degraded", "shadow_id": "uuid", "reason": "source_timeout", "ts": "..."}
{ "event": "shadow_reconnected", "shadow_id": "uuid", "ts": "..."}
}
```

25.5 CRDT Federation Sync Stream

Attribute	Value
URL	wss://{grid_host}/ws/v1/crdt/federation/{peer_id}
Auth	mTLS (internal service mesh only)
Purpose	Real-time CRDT gossip stream between two Grid federation nodes

```
// Messages (bidirectional):
// Batch of CrdtOperations from peer:
{ "op": "crdt_op_batch", "peer_id": "uuid", "batch_id": "uuid", "ops": [ CrdtOperation, ... ], "vector_clock": { "node-id": 42 }, "ts": "..."}

// Acknowledge receipt of a batch:
{ "op": "ack", "batch_id": "uuid" }

// VectorClock heartbeat exchange:
```



```
{ "op": "heartbeat", "vector_clock": { "node-id": 42 }, "ts": "..." }

// Schema delta (new dimension or attr key):
{ "op": "schema_delta", "cube_id": "uuid",
  "events": [ SchemaEvent, ... ], "ts": "..." }

// Conflict surfaced during apply:
{ "op": "conflict_surfaced", "conflict": ConflictRecord, "ts": "..." }
```

Part XIV — Server-Sent Events (SSE)

26. SSE Endpoints

Server-Sent Events provide a lightweight, one-directional streaming alternative to WebSocket. Use SSE for read-only monitoring, dashboard feeds, and integration with systems that consume SSE natively (e.g. browser EventSource API, some serverless environments).

GET	/api/v1/cubes/{cube_id}/events
-----	--------------------------------

Subscribe to a real-time SSE stream of cell mutations for a Hypercube.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.
entity_id	query	uuid	No	Filter to a specific HyperRow.
attr_key	query	string	No	Filter to a specific attribute key.
event_types	query	string	No	Comma-separated event types to receive.

Responses:

Status	Meaning	Response Body
200 OK	Streaming	Content-Type: text/event-stream. SSE event stream.
403	Insufficient scope	

Note: SSE connections are kept alive with comment heart beats every 30s. The Last-Event-ID

Status	Meaning	Response Body
header is supported for reconnection — the server will replay missed events from the specified event ID.		

GET	/api/v1/spaces/{space_id}/events
-----	----------------------------------

Subscribe to a real-time SSE stream of Space activity events.

Parameters:

Name	In	Type	Required	Description
space_id	path	uuid	Yes	Space identifier.

Responses:

Status	Meaning	Response Body
200 OK	Streaming	Content-Type: text/event-stream.

```
// SSE event format:
id: evt-uuid-123
event: cell_updated
data: {"cube_id":"uuid","dl_key":"uuid","attr_key":"health_score","new_value":87.4,"ts":"..."}

id: evt-uuid-124
event: row_created
data: {"cube_id":"uuid","dl_key":"new-entity-uuid","ts":"..."}
```

```
: heartbeat 2026-03-22T10:00:30Z  
  
// Reconnection: client sends Last-Event-ID header = "evt-uuid-124"  
// Server replays all events since that ID (up to 1000 events, 24h max window)
```

Part XV — HypercubeView API

27. Views

GET	/api/v1/cubes/{cube_id}/views
-----	-------------------------------

List all HypercubeViews for a cube. Views with visibility=Public or SpaceShared are visible to all; Private views only to their creator.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.
visibility	query	string	No	Private Shared SpaceShared Public Template — filter.
render_mode	query	string	No	Filter by render mode.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ data: HypercubeView[], meta: PaginationMeta }

POST	/api/v1/cubes/{cube_id}/views
------	-------------------------------

Create a new HypercubeView.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.

Request Body:

```
{
  "name": "Q2 Active Projects by Health",
  "primary_axis": "entity_axis_id",           // D1 → rows
  "secondary_axis": "property_axis_id",       // D2 → columns
  "dim_slices": [                             // filter on D3+ axes
    { "axis_id": "time_axis_id", "op": "eq", "value": "2026-Q2" }
  ],
  "dim_folds": [],
  "dim_expands": [],
  "attr_schema": [                            // column order and display
    { "attr_key": "name", "width": 300, "pinned": true },
    { "attr_key": "status", "width": 120 },
  ]
}
```

```
    { "attr_key": "health_score", "width": 100 },
    { "attr_key": "budget_remaining", "width": 120 }
  ],
  "filters": [
    { "attr_key": "status",      "op": "eq", "value": "Active" },
    { "attr_key": "item_type",   "op": "eq", "value": "Project" }
  ],
  "sorts": [
    { "attr_key": "health_score", "direction": "asc" }
  ],
  "groups": [],
  "render_mode": "Grid2D",
Grid2D|Kanban|Gantt|Calendar|Timeline|
HierarchyTree|Treemap|NetworkGraph|
LinkForest|PivotTable|HeatMap|Matrix|
//      N-DimExplorer|Custom(plugin_id)
  "highlight_rules": [
    { "condition": "health_score < 50", "color": "#FEE2E2", "label": "At Risk" }
  ],
  "shadow_policy": "Include",
  "graph_overlay": false,
  "ai_overlay": true,
  "visibility": "SpaceShared"
}
```

Responses:

Status	Meaning	Response Body
201 Created	View created	HypercubeView object
400	Invalid render mode	Render mode not compatible with specified axes.
403	Insufficient tier	Requires Editor+ tier.

GET	/api/v1/cubes/{cube_id}/views/{view_id}/rows
-----	--

Render rows for a HypercubeView with all view configuration applied: DimSlices, DimFolds, DimExpands, filters, sorts, groups, shadow policy, and highlight rules.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.
view_id	path	uuid	Yes	HypercubeView identifier.
identity_tag	query	string	No	Override identity partition for this render.
as_of	query	datetime	No	Time-travel render.
limit	query	integer	No	Default: 50. Max: 500.
cursor	query	string	No	Pagination cursor.

Responses:

Status	Meaning	Response Body
200 OK	Success	{ view_id, columns: [{ attr_key, display_name, width, pinned }], rows: [{ d1_key, attrs, highlight_color?, is_shadow?, group_label? }], groups?: [...], summary_row?: {...}, meta: PaginationMeta }
404	View not found	

Note: Group headers are included as special rows with group_label and aggregated values. The summary_row (if configured) appears as the first row. ShadowRows are marked with is_shadow=true and the source

Status	Meaning	Response Body
grid/identity.		

PATCH	/api/v1/cubes/{cube_id}/views/{view_id}
-------	---

Update a HypercubeView configuration.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.
view_id	path	uuid	Yes	HypercubeView identifier.

Request Body:

<pre>{ "name": "Q2 Active Projects by Risk", "sorts": [{ "attr_key": "risk_score", "direction": "desc" }], "visibility": "Public" }</pre>

Responses:

Status	Meaning	Response Body
200 OK	Updated	Updated HypercubeView
403	Insufficient tier	Requires Editor+ tier. Updating visibility to Public requires Manager+.

DELETE	/api/v1/cubes/{cube_id}/views/{view_id}
--------	---

Delete a HypercubeView. Does not affect any cell data.

Parameters:

Name	In	Type	Required	Description
cube_id	path	uuid	Yes	Hypercube identifier.
view_id	path	uuid	Yes	HypercubeView identifier.

Responses:

Status	Meaning	Response Body
204 No Content	Deleted	No body.
403	Insufficient tier	Requires Owner of this view.

Part XVI — Rate Limits, Resilience Patterns, and SDK Guide

28. Rate Limits

Endpoint Group	Default Limit	Burst	Headers
All REST endpoints	1,000 req/min per identity	200 req/10s	X-RateLimit-Limit, X-RateLimit-Remaining, X-RateLimit-Reset
POST /cubes/{id}/rows	500 req/min per identity	100 req/10s	Same headers
PATCH /cubes/{id}/rows/{id}	500 req/min per identity	100 req/10s	Same headers
POST /cubes/{id}/query	200 req/min per identity	50 req/10s	Same headers
POST /query/async	50 req/min per identity	10 req/10s	Same headers
POST /query/nl	60 req/min per identity	15 req/10s	Also: X-AI-Credit-Remaining
POST /ai/compute	200 req/min per identity	50 req/10s	Also: X-AI-Credit-Remaining
POST /graph/edges	200 req/min per identity	50 req/10s	Same headers
WebSocket connections	10 concurrent per identity	—	Connection rejected with 429 if limit exceeded
Federation sync endpoints	No limit (internal)	—	mTLS only — not accessible externally

Note: Rate limits can be increased via plan upgrades. Contact support for enterprise rate limit customization. WritebackService (AI engine writeback) is excluded from per-identity rate limits — it uses per-engine quotas.

29. Resilience Patterns for API Consumers

Pattern	When to Apply	Implementation Guidance
Idempotency Keys	All POST/PUT/PATCH requests that create or mutate data	Generate a UUID v4 per operation attempt. Set X-Idempotency-Key header. On network failure, retry with the same key. The server stores results for 24 hours — duplicate requests return the same response.
Exponential Backoff	429 (rate limit), 503 (unavailable), 502 (upstream error)	Initial delay: 1s. Factor: 2. Max delay: 60s. Add $\pm 10\%$ jitter. Do NOT retry on 400, 401, 403, 404, 409.

Pattern	When to Apply	Implementation Guidance
Optimistic Concurrency	PATCH operations on frequently-updated rows	Include If-Match header with current ETag from GET response. On 409 version_conflict: re-fetch, re-apply your change, retry PATCH with new ETag.
Read-Your-Writes	After a write, if you immediately read the same entity	The server guarantees read-your-writes for 5 seconds by routing to the same PostgreSQL primary. For guaranteed consistency: include X-After-Write: {vector_clock} header on subsequent reads.
Circuit Breaking	Calling Hypergrid from a service with SLA dependencies	Trip circuit at 50% error rate over 10s. Half-open state: allow 1 probe request every 30s. On trip, serve cached or degraded response.
Streaming Large Queries	Any query expected to return >10,000 rows	Use POST /query/async and stream results via GET /query/async/{id}/stream. Set Accept: application/x-ndjson for line-by-line processing.
Webhook Delivery	Receiving real-time updates in server-side integrations	Register webhooks via POST /webhooks. The server will POST events with HMAC-SHA256 signatures. Validate X-Hypergrid-Signature header on receipt. Respond with 200 within 5s. Failed webhooks retried 5 times with exponential backoff.
Cache Key Strategy	Client-side caching of HyperRow data	Cache key: {cube_id}:{d1_key}:{attrs_hash}. Invalidate on: ETag change (compare X-Entity-Version header), WebSocket cell_updated event, or TTL expiry. Recommended TTL: 60s for Tier-2 AI attrs, 5s for LWW scalar attrs.

30. SDK Quick Reference

30.1 TypeScript / JavaScript SDK

```
import { HypergridClient, HyperQL } from '@hypergrid/sdk';

const client = new HypergridClient({
  baseUrl: 'https://kogi.hypergrid.io/api/v1',
  apiKey: process.env.HYPERGRID_API_KEY,
  gridId: 'kogi-production-grid-id',
});

// List HyperRows with DimSlice filter
const rows = await client.cubes('kogi.portfolio.components').rows.list({
  filter: "D2='status' AND value='Active' AND D2='health_score' AND value < 70",
  attrs: ['name', 'status', 'health_score', 'budget_remaining'],
  sort: 'health_score:asc',
  limit: 50,
});

// Read a single HyperRow
const component = await
client.cubes('kogi.portfolio.components').rows.get(entityId);

// Write attributes (CRDT-safe)
```

```

await client.cubes('kogi.portfolio.components').rows.update(entityId, {
  attributes: {
    name: 'Renamed Project',
    status: { from: 'Active', to: 'Deprecated' }, // Lattice transition
    tags: { add: ['q2-2026'], remove: ['q1-2026'] } // OR-Set
  }
});

// Execute HyperQL
const result = await client.query(
  HyperQL.select('D1.entity_id', 'name', 'health_score')
    .from('kogi.portfolio.components')
    .where("status = 'Active' AND health_score < 70")
    .orderBy('health_score', 'asc')
    .limit(20)
);

// Subscribe to real-time updates (WebSocket)
const sub = client.cubes('kogi.portfolio.components').subscribe({
  scope: 'attr', attrKey: 'health_score',
  onUpdate: (event) => { console.log('Health score updated:', event); }
});

// Natural language query
const nlResult = await client.ai.nlQuery({
  question: 'Which projects are over budget this quarter?',
  cubeContext: ['kogi.portfolio.components'],
});
console.log(nlResult.generatedHyperql); // transparency: show generated HyperQL

```

30.2 Python SDK

```

from hypergrid import HypergridClient, HyperQL, DimSlice

client = HypergridClient(
    base_url="https://kogi.hypergrid.io/api/v1",
    api_key=os.environ["HYPERGRID_API_KEY"],
    grid_id="kogi-production-grid-id",
)

# List HyperRows
rows = client.cubes["kogi.portfolio.components"].rows.list(
    filter=DimSlice.where("status").eq("Active").and_("health_score").lt(70),
    attrs=["name", "status", "health_score"],
    limit=50,
)

# Create a HyperRow
new_row = client.cubes["kogi.portfolio.components"].rows.create({
    "name": "New Project",
    "status": "Draft",
    "tags": ["q2-2026", "engineering"],
    "budget": {"amount": "10000.00", "currency": "USD"},
})

# Execute async HyperQL
job = client.query.async_execute(
    "SELECT D1.entity_id, FOLD D3 WITH SUM(value) AS total "

```

```

    "FROM kogi.portfolio.finances "
    "WHERE D2.metric_name = 'revenue' "
    "GROUP BY D1.entity_id ORDER BY total DESC",
    output_format="parquet"
)
# Stream results as Parquet
with open("results.parquet", "wb") as f:
    for chunk in job.stream():
        f.write(chunk)

# Graph traversal
tree = client.graph.nodes[node_id].tree(max_depth=3, edge_type="Hierarchy")
print(f"Found {len(tree.nodes)} nodes in hierarchy")

# Time-travel query
historical = client.cubes["kogi.portfolio.components"].rows.list(
    as_of="2026-01-01T00:00:00Z",
    filter=DimSlice.where("status").eq("Active"),
)

```

30.3 Rust SDK

```

use hypergrid_sdk::{HypergridClient, HyperQL, DimCoordinate, CrdtOp};

let client = HypergridClient::builder()
    .base_url("https://kogi.hypergrid.io/api/v1")
    .api_key(std::env::var("HYPERGRID_API_KEY")?)
    .build()?;

// Get a specific HyperCell (N-dim coordinate)
let coord = DimCoordinate::new(vec![
    DimKey::Uuid(entity_id),
    DimKey::Text("health_score".into()),
    DimKey::Text("2026-Q2".into()), // D3=TimeAxis
    DimKey::Text("EMEA".into()),   // D4=GeoAxis
]);
let cell = client.cubes("qala.solutions.metrics").cells().get(&coord).await?;

// Write with CRDT operation
client.cubes("kogi.portfolio.components")
    .cells()
    .patch_attr(&entity_coord, "status", CrdtOp::TransitionState {
        from: "Active".into(),
        to: "Deprecated".into(),
    })
    .await?;

// Subscribe to real-time updates (async stream)
let mut stream = client
    .cubes("kogi.portfolio.components")
    .rows()
    .subscribe_row(entity_id)
    .await?;

while let Some(event) = stream.next().await {
    match event {
        CellEvent::Updated { attr_key, new_value, .. } => {
            println!("Attr '{}' updated to {:?}", attr_key, new_value);
        }
    }
}

```

```
    }  
    CellEvent::AiWriteback { engine, attr_key, value, confidence } => {  
        println!("AI engine '{}' updated '{}'={:?} (conf={})",  
            engine, attr_key, value, confidence);  
    }  
    _ => {}  
}  
}
```

Part XVII — Performance Targets and Appendices

31. API Performance Targets

Operation	Target p50	Target p99	Notes
GET /cubes/{id}/rows (50 rows)	< 30ms	< 150ms	Single cube, 2D dense, no fold. Includes computed Tier-1 attrs. Redis cache serves AI attrs.
GET /cubes/{id}/rows (500 rows)	< 100ms	< 400ms	Larger page size with sort. Index-assisted sort on scored columns.
GET /cubes/{id}/rows/{d1_key}	< 5ms	< 20ms	Single HyperRow primary key lookup. Redis serves AI attr cache.
PATCH /cubes/{id}/rows/{d1_key}	< 10ms	< 50ms	Single entity, 1–5 attr updates. CrdtLog Redis write + async PG flush.
POST /cubes/{id}/query (small, < 1K rows)	< 100ms	< 500ms	Simple DimSlice query, no fold. PostgreSQL index-assisted.
POST /cubes/{id}/fold (1M rows)	< 2s	< 8s	Single DimFold on N=3 cube. ClickHouse backend for > 100K rows.
POST /query/nl (NL → HyperQL + execute)	< 2s	< 8s	NL translation + HyperQL execution. Depends on AI engine latency.
GET /ns/resolve (Redis hit)	< 2ms	< 8ms	Redis-cached namespace resolution.
GET /ns/resolve (PG fallback)	< 10ms	< 40ms	Cache miss — PostgreSQL index lookup.
POST /graph/edges (no consent)	< 15ms	< 60ms	Edge creation without consent flow.
GET /graph/nodes/{id}/tree (depth=3)	< 100ms	< 500ms	BFS on PostgreSQL adjacency list. Cached for hot nodes.
GET /graph/path (6 hops)	< 200ms	< 800ms	Dijkstra on PG with adjacency index.
POST /cubes/{id}/snapshot	< 5s	< 30s	Snapshot large cube. Async — 202 returned immediately.
POST /cubes/{id}/restore/{snap_id}	< 30s	< 5min	Full cube restore. Async — 202 returned immediately.
WebSocket cell_updated broadcast (100 clients)	< 30ms	< 150ms	Redis pub/sub fanout. Measured end-to-end from write commit.
Shadow sync delta (MirrorAttribute)	< 500ms	< 2s	Kafka event → consumer → PG write → WebSocket push.

32. Appendix A — AttributeType Reference

AttributeType	JSON Representation	Default CRDT	Notes
Text	string	LWW	UTF-8 string. Max 65,535 bytes.
Number	number (float64)	LWW	IEEE 754 double. Use Currency for monetary values.
Integer	number (integer)	LWW	64-bit signed integer.
Currency	{ amount: string, currency: string }	LWW	Decimal amount string (no float imprecision). ISO 4217 currency code.
Percent	number (0.0–100.0)	LWW	Stored as float. Rendered as percentage.
Bool	boolean	LWW	true false.
Date	string (YYYY-MM-DD)	LWW	ISO 8601 date only. No timezone.
DateTime	string (ISO 8601)	LWW	UTC timestamp. Stored with nanosecond precision.
Duration	number (seconds)	LWW	Duration in seconds. Rendered as human-readable (2h 30m).
Enum	string	LWW	One of a declared variant set. Validated at write time.
MultiEnum	string[]	OR-Set	Multiple enum variants. OR-Set: concurrent adds survive.
Relation	{ cube_id: uuid, d1_key: uuid }	LWW	Reference to one HyperRow in any cube.
MultiRelation	{ cube_id, d1_key }[]	OR-Set	References to multiple HyperRows.
User	uuid (identity_id)	LWW	Reference to an identity in the Identity system.
Tag	string	LWW	Single tag string.
TagSet	string[]	OR-Set	Set of tags. OR-Set: concurrent tag additions both survive.
Json	any JSON value	LWW (or JsonCRDT)	Arbitrary JSON. Use JsonCRDT semantics for deep-merge behavior.
Computed	matches output attr_type	LWW (system)	Tier-1 formula result. System-write on read evaluation.
AI	matches output attr_type	LWW (system)	Tier-2 AI signal. Written by AIEngineAdapter via WritebackService.
Formula	{ expression: string, result: any }	LWW (system)	Stored formula expression + last computed result.
Audit	{ actor, action, ts }	AppendOnlyLog	Append-only audit record. System-managed.
Custom	any JSON	Plugin-defined	Registered via HypercubePlugin AttributeTypePlugin interface.

33. Appendix B — EdgeType Quick Reference

EdgeType	Consent?	Shadow?	Cycle Detection?	CRDT	Notes
Hierarchy	No	No	Yes	OR-Set (edges)	Parent-child. Enables ROLLUP/DRILLDOWN. Max depth enforced by configured limit.
Dependency	No	Optional	Yes	OR-Set (edges)	Blocking. from_node cannot complete before to_node.
Association	No	No	No	OR-Set (edges)	Soft lateral reference. No blocking semantics.
Contains	No	No	No	OR-Set (edges)	Ownership containment.
Derives	No	Optional	No	OR-Set (edges)	Versioning / fork relationship.
CrossGridLink	Yes	Yes	No	Consent: LWW	Inter-Grid connection. Core link network atom.
ShadowOf	System	N/A	No	System	System edge. Inverse of CrossGridLink. Auto-created on consent.
SpaceMembership	Space Type	No	No	OR-Set (members)	Identity → Space. Role on edge attrs.
Collaborates	Mutual	Yes	No	Consent: LWW	Bidirectional. Both parties consent.
References	No	No	No	OR-Set (edges)	Citation/reference. Lighter than Dependency.
FederationPeer	Grid-trust	System	No	System	Grid-to-Grid CRDT federation link.
NamespaceAliases	No	No	No	LWW	Namespace path alias/redirect.
ComputedFrom	No	No	No	System	Provenance / lineage edge.
SubscribesTo	No	Optional	No	OR-Set (subs)	Change notification subscription.
InvestedIn	Yes	Yes	No	Consent: LWW	Economic investment. Platform-specific semantics.
Employs	Yes	Yes	No	Consent: LWW	Employment/contracting. Worker consents.
Custom(String)	Plugin	Plugin	Plugin	Plugin	Custom edge type registered via HG-PLUGIN EdgeTypePlugin.

34. Appendix C — CRDT Semantics Quick Reference

CrdtSemantics	Merge Rule	REST write body	EventLog Op	Suitable For
LastWriteWins	Higher VectorClock timestamp wins. Tiebreak: ActorId lex sort.	{ value: ... }	SetAttr	All scalar fields: Text, Number, Enum, Bool, Date, Json, references
OR-Set	All concurrent Adds survive. Tagged Removes only remove their exact entry.	{ add: [...], remove: [...] }	AddToSet / RemoveFromSet	TagSet, MultiEnum, MultiRelation, owner lists, child ID sets
GrowOnlyCounter	Sum of all increments. Never decrements.	{ incr: N } (N must be > 0)	IncrCounter	view_count, like_count, follower_count, restart_count
PNCounter	Sum of (pos_ops – neg_ops) per node.	{ incr: N } (N any sign)	IncrCounter	budget_spent, hours_logged (correctable accumulators)
MaxRegister	max(value_a, value_b).	{ value: N }	SetAttr	version numbers, sequence numbers, generation counters
MinRegister	min(value_a, value_b).	{ value: N }	SetAttr	earliest_deadline, creation timestamps
Lattice	join(state_a, state_b) per configured partial order. No backward transitions.	{ from: state, to: state }	TransitionState	lifecycle_state, ComponentStatus, SolutionLifecycle, CCR status
AppendOnlyLog	All appends survive. Causally ordered by VectorClock.	{ append: value }	AppendLog	Comment threads, change_history, governance_audit
JsonCrdt	Deep-merge: recursive LWW per leaf JSON key.	{ deep_merge: { key: value } }	DeepMergeJson	JSON configuration blobs requiring partial-key updates
Custom	Plugin provides merge(a, b, ctx) → value.	Plugin-defined	Plugin-defined	Domain-pack fields, third-party attribute types

35. Appendix D — PermissionTier Reference

Tier	Integer	Label	Can Write	Kogi	Ume	Qala
Public	0	Public/anonymous	Only public-visibility attrs on read-only endpoints	Anonymous read	Anonymous read	Public solution browsing
Viewer	1	Viewer	No write access	Follow/watch portfolio	Employee read-only	SDE observer
Subscriber	1	Subscriber	No write access	Subscribe to updates	Subscribe to alerts	Release notifications

Tier	Integer	Label	Can Write	Kogi	Ume	Qala
Member	2	Member	Own standard fields	Portfolio collaborator	Org member	SDE team member
Contributor	3	Contributor	Content fields on assigned rows	Portfolio contributor	Record submitter	Junior developer
Editor	4	Editor	All content + config + timeline + budget fields	Portfolio editor	Domain record editor	Developer with SDE access
Manager	5	Manager	All Editor + governance + member roster + Space settings	Portfolio manager	Department head	Tech Lead / Release Manager
Owner	6	Owner	All fields including governance records + archive	Full portfolio ownership	System Administrator	Factory Admin
Admin	7	Admin (platform)	All fields + platform override	Platform Admin	Ume Kernel Admin	Qala Platform Admin
System	8	System (internal)	AI-computed attrs + federation sync + system metadata	AI engine / sync agent	Kernel services	AI Agent / CI pipeline

End of Document

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